

CLEAN ENERGY
TECHNOLOGY
OBSERVATORY

Hydropower and Pumped Storage Hydropower in the European Union

*Status Report on Technology Development,
Trends, Value Chains and Markets*

2025

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Abstract

In 2022, the European Commission established the Clean Energy Technology Observatory (CETO) to address the complex transition to a climate-neutral society, emphasizing advanced technologies and innovation in sectors like hydropower. The energy crisis, brought on by the war against Ukraine, has highlighted the key role of hydropower in providing grid stability and dispatchable generation, as well as additional water supply and grid services. Pumped-Storage Hydropower holds more than 90% of energy storage, while generation from traditional hydropower provided 10% of total EU's electricity generation. Increasing investments in the operation of hydropower plants are essential to bear the significant increase of stop&start cycles caused by the volatility of wind and photovoltaic electricity generation, and to provide flexibility to the energy system.

Hydropower is the flagship of renewables in the EU: the EU hydropower fleet is mostly "*made in the EU*", with a centenary history. EU hydro companies contribute to global exports with the largest share and own the largest share of high-value inventions among the top 10 world companies. Hydropower reservoirs also provide services in the integrated water management sector, spanning from flood control to water supply in dry periods, storing the largest amount of water among all the types of water reservoirs in the EU. Water storage is a key option to grapple with climate changes and increased misalignment between water demand and availability, as also mentioned in the new EU Water Resilience Strategy. However, dams in freshwater and coastal water systems can cause environmental damages and require large initial investment costs. Therefore, considering the far-reaching effects across the Water-Energy-Society-Ecosystem nexus dimensions, sustainable hydropower needs to achieve a good balance between electricity generation and electricity grid services, impacts on ecosystems and benefits for society, supporting the achievement of the Green Deal targets and the objectives of renewable energy and water/environmental policies. Several sustainable hydropower options exist in the EU, that do not increase river fragmentation and minimize adverse impacts, and that are summarized in this report.

Hydropower remains a key technology to strengthen the competitiveness of the EU in an increasingly challenging world and to support the green energy transition.

Foreword on the Clean Energy Technology Observatory

The European Commission set up the Clean Energy Technology Observatory (CETO) in 2022 to help address the complexity and multi-faced character of the transition to a climate-neutral society in Europe. The EU's ambitious energy and climate policies create a necessity to tackle the related challenges in a comprehensive manner, recognizing the important role for advanced technologies and innovation in the process.

CETO is a joint initiative of the European Commission Joint Research Centre (JRC), who run the observatory, and Directorate Generals Research and Innovation (R&I) and Energy (ENER) on the policy side. Its overall objectives are to:

- monitor the EU research and innovation activities on clean energy technologies needed for the delivery of the European Green Deal
- assess the competitiveness of the EU clean energy sector and its positioning in the global energy market
- build on existing Commission studies, relevant information & knowledge in Commission services and agencies, and the Low Carbon Energy Observatory (2015–2020)
- publish reports on the Strategic Energy Technology Plan ([SET-Plan](#)) SETIS online platform

CETO provides a repository of techno- and socio-economic data on the most relevant technologies and their integration in the energy system. It targets in particular the status and outlook for innovative solutions as well as the sustainable market uptake of both mature and inventive technologies. The project serves as primary source of data for the Commission's annual progress reports on [competitiveness of clean energy technologies](#). It also supports the implementation of and development of EU research and innovation policy.

The observatory produces a series of annual reports addressing the following themes:

- Clean Energy Technology Status, Value Chains and Market: covering advanced biofuels, batteries, bioenergy, carbon capture utilisation and storage, concentrated solar power and heat, geothermal heat and power, heat pumps, hydropower & pumped hydropower storage, novel electricity and heat storage technologies, ocean energy, photovoltaics, renewable fuels of non-biological origin (other), renewable hydrogen, solar fuels (direct) and wind (offshore and onshore).
- Clean Energy Technology System Integration: building-related technologies, digital infrastructure for smart energy system, industrial and district heat & cold management, standalone systems, transmission and distribution technologies, smart cities and innovative energy carriers and supply for transport.
- Foresight Analysis for Future Clean Energy Technologies using Weak Signal Analysis
- Clean Energy Outlooks: Analysis and Critical Review
- System Modelling for Clean Energy Technology Scenarios
- Overall Strategic Analysis of Clean Energy Technology Sector

More details are available on the [CETO web pages](#)

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Executive Summary

EU objectives and challenges

In the European Union (EU), the hydropower installed capacity in 2023 was 153 GW and generated approx. 300 TWh, which is the second highest share from renewable energy sources, after wind energy. Ninety percent of hydropower plants are small (<10 MW), contributing up to 10% of the total installed capacity. Currently, the EU hosts 46 GW of PSH (turbine's power capacity), which is a quarter of the global installed capacity (50.9 GW in China); in 2023, PSH consumed (pumping mode) and generated approx. 40 TWh and 45 TWh, respectively. PSH allows to integrate the volatile energy production from non-dispatchable energy sources (e.g. wind and photovoltaic power plants). In the last decade in the EU, 6 GW of new hydropower have been added, and several GW are estimated to have been refurbished.

Hydropower is a renewable and flexible energy source that contributes to a resilient energy system, ensuring grid stability (preventing blackouts) and ancillary services. It plays a key role in the long-term decarbonisation scenarios, contributing to reach the renewable energy targets set in the Renewable Energy Directive (Directive 2009/28/EC) and REPowerEU. However, stop&start cycles, caused by the volatility of photovoltaic and wind energy production, are constantly increasing, challenging hydropower operators and the equipment. Multi-purpose hydropower projects and large dams/reservoirs can play important additional functions for society, providing water storage for irrigation and water provision, flood and drought risk management, river navigation and recreation. The need of water storage for integrated water management is increasing due to climate changes and is also supported by the objectives of the Water Resilience Strategy. On the other hand, hydropower barriers installed in freshwater and coastal water systems are a source of impact on aquatic environment. These impacts are regulated in the Water Framework Directive (Directive 2000/60/EC), which is aimed at the preservation or recovery of the "good ecological status"^a of the water bodies. New barriers can interrupt river continuity (i.e. migration of fish and other aquatic species), reservoirs generate impoundment, with consequent hydrological and morphological alterations, and hydropower turbines may cause damages to fish. In 2020, the AMBER project has identified at least 450,000 barriers in EU's rivers (225,000 are dams and weirs), while the number of hydropower plants in the EU is estimated to be approximately 25,000.

Technology status

Technical maturity is well established, with overall power plant's efficiencies generally exceeding 80%, and that can reach up to 90%. IRENA's and World Bank's analysis identified hydropower as currently one of the least expensive forms of renewable electricity generation when looking at the Levelized Cost of Electricity, LCOE, despite high initial investment costs. Despite this, challenges, innovation needs and emerging technologies have not disappeared.

Investment and funding

Hydropower contribution to the EU annual gross domestic product (GDP) is approx. EUR 25 Billion considering electricity generation, but if the multi-services associated to hydropower are considered, this value may increase of tens of Billion. In the last decade (2014-2023), public spending for R&D on hydropower in the EU ranged between EUR 8.4 Million and EUR 22.5 Million per year (2024 data not complete yet), with decreasing trends after peaks. Within this context, the European Commission (EC) supported and supports several research projects (on average, EUR 10 Million per year under the Horizon Europe program) that have generated an important scientific, industrial and technical impact worldwide, with EU technologies that can generate a large opportunity for export. In terms of scientific publications, the EU is in second position, after China. Although China is the main patent leader (partially also due to the different patenting procedure in the country), the EU holds 16% of

^a Ecological status describes the structure and functioning of aquatic ecosystems associated with surface waters. The good status shows for the aquatic environment low levels of distortion resulting from human activity.

all high-value inventions globally (2020-2022), 41% of its inventions are high-value, and 49% of high-value inventions from the top 10 global leaders is owned by EU companies.

Value chain

Employment in hydropower industry spans various value chain elements, including project design, manufacturing, project construction and operations and maintenance (O&M). The sector's employment generally includes engineers, geologists, ecologists, economists, technicians, and skilled workers. It also provides employment to scientists working in corporate and academic R&D activities. IRENA estimates that every 1 MW of community-owned hydropower installed generates ten full-time equivalent jobs in every year of its operation. This is significantly more than any other renewable energy technology. In 2023 there were 102,000 employed people in the manufacturing, construction and R&D in the EU's hydropower sector, with Austria, Germany, France and Italy, mostly located at the heart of the Alps, hosting the largest numbers of employees, but without counting all the associated engineering, financing and environmental services associated, especially during the project planning and design.

Sustainability

As the crosscutting effects of hydropower, and its interactions with the Earth's spheres, are quite far-reaching, hydropower is a complex and challenging sector within the Water-Energy-Society-Ecosystem (WEFE) nexus (SWOT in Table 1). Therefore, sustainable hydropower needs to achieve a good balance between clean electricity generation and impacts (and benefits) on the environment and on the society. In the EU, an untapped sustainable hydropower potential still exists, e.g. of new PSH typologies, modernization of the existing ones, digitalisation of the operation and small hydropower development in existing infrastructures ("hidden hydropower"). Hybridisation with other energy technologies, e.g. floating photovoltaics on hydropower reservoirs, hybridization with batteries and waste-heat recovery are additional opportunities. Ocean (tidal and wave) power plants use turbines adapted from the hydropower sector. Lifespan of hydropower is longer than other renewables and does not rely on critical materials.

EU positioning and global competitiveness

Global exports of turbines and parts rose from EUR 791 Million in 2023 to EUR 932 Million in 2024, with EU exports (including intra-EU trade) increasing from EUR 351 Million to EUR 452 Million. Over 2022-2024, the EU accounted for 47% (49% over 2020-2022) of global exports (including intra-EU trade), while extra-EU exports (excluding intra-EU trade) represented 38% of global transactions (42% over 2021-2023). The EU met 80% of its import needs of turbines and parts through intra-EU trade. However, the real dimension of the hydropower sector is much larger than just the dimension associated to the turbine. If all the technologies associated to the hydropower chain were considered (approx. 150 types, e.g. including pipes, penstocks, valves, gates), the European House Ambrosetti found that annual production would be EUR 27.7 Billion and EUR 43.4 Billion for Italy and Germany, respectively, instead of EUR 107 Million and EUR 150 Million in 2022, respectively. According to this study, in 2022 the total EU's Production value associated to hydropower was EUR 132.3 Billion from the first ten EU member states, instead of EUR 562 Million. Furthermore, a considerable proportion of investments in the hydropower sector refers to the civil works and associated consultancy services that are very difficult to track. The large European hydropower operators continue to invest in many hydropower and dam projects outside of Europe.

Challenges and directions

The benefits of hydropower are challenged by financial, regulatory and environmental challenges. The declining revenues from the provision of the ancillary services are not enough to cover the fixed investment and administrative costs. Renewable energy policy attention in the past two decades has focused primarily on wind and solar PV technology expansion (and lowering their cost), with less focus on the key support of hydropower, with its flexibility and its "made in EU" dimension. Multiple water supply services are often not officially recognized. Hydropower projects have longer pre-

development, construction and operational timelines than other renewable energy technologies. Investment risks for large hydropower schemes are higher, requiring specific policy instruments and incentives, as well as a longer-term policy perspective and vision. These challenges should serve as a catalyst for a more comprehensive dialogue among stakeholders (e.g. industry, academy, associations, citizens and governmental institutions). To gain wide acceptance and to obtain a win-win solution among all stakeholders' needs, the large hydropower infrastructures have to be designed as multi-purpose projects by multidisciplinary teams with a complex system approach. The available sites for new large hydropower schemes in the EU are limited, due to environmental and economic issues, while the EU leadership is challenged by global competitiveness. The abovementioned challenges require excellence in engineering sciences, environmental management and new innovative planning approaches.

This report updates the [2024 report](#).

SWOT analysis

Table 1. CETO SWOT analysis for the competitiveness of the hydropower sector (focus on the EU).

Strength - Mature technology, high efficiency - High flexibility, ensuring supply during peak demand - Grid stabilizer - Long lifespan - Storage capacity (water and energy) of reservoirs - Multi-purpose benefits of reservoirs and dams - The lowest Ozone Layer Depletion indicator, the highest Energy Returned on Energy Invested ratio among electricity technologies, the lowest pressure on mineral resources and amongst the lowest water footprints during construction and manufacturing per GWh - The EU is a leader in scientific research, technological innovation, export - Hydro turbines and generators can play an important role in the grid frequency stability	Weakness - Environmental and social impacts associated with the construction of new barriers (and dams), and the mitigation of the existing ones, demand significant and costly mitigation measures (e.g. habitat, fish migration, hydropеaking, hydro-morphological alterations). In reservoirs: sedimentation and impoundment. - Limited potential for large hydropower capacity expansion - Long construction periods and large investment needs/risks of large power plants - Additional benefits (water supply and grid services) are not all remunerated, and often are not recognized
Opportunities - Hidden potential in existing facilities in the water sector - Attractive for rural and decentralized electrification in developing countries - Integration of intermittent renewable energy sources with pumped-storage hydropower - Modernization of hydropower infrastructure is needed in the EU and can bring additional benefits in light of climate change and water resilience - Still some potential for increasing pumped-storage hydropower - Potential for hybridization with other energy technologies - New reservoirs in high altitude using new lakes created by glacier retreat - Export potential of electro-mechanical equipment - Increase of exports with "made in EU" products in countries where hydro is emerging	Threats - Substantial uncertainties in long approval processes - Climate changes may increase or decrease water availability and its yearly distribution - Loss of reservoirs' volume due to sedimentation - In the EU, need of improved market rules and to remunerate the additional benefits - Loss of knowledge due to low attraction of traditional engineering fields for young professionals - Moderate public awareness on the benefits of hydropower - Decrease of exports outside of the EU due to low support to manufacturing with respect to other competitive countries - The centenary brand, widespread throughout the world, "made in EU" is threatened

1 Introduction

1.1 Scope and context

This report on Hydropower and Pumped-Storage Hydropower (PSH) is part of the annual series of reports from the Clean Energy Technology Observatory (CETO). This report builds on previous EU studies in this field and updates the [2024 CETO report](#). It provides an overview of the current state of Hydropower and Pumped-Storage Hydropower, including technology maturity status, development and trends, value chain analysis, global market and EU positioning.

Hydropower and Pumped-Storage Hydropower play a crucial role in achieving the objectives of the European Green Deal and EU policies or initiatives, e.g., REPowerEU, Net Zero Industry Act. The EU has set ambitious targets for renewable energy and carbon neutrality, and Hydropower and Pumped-Storage Hydropower is expected to contribute significantly to these efforts. Furthermore, the water storage capacity of hydropower reservoirs can also play a key role in climate change adaptation and in the Water Resilience Strategy. Small hydropower leveraging hydraulic infrastructures is also supported by other recent EU policy programmes. The refurbishment of SHP is eligible under the Social Climate Fund (SCF); the Water Resilience Strategy aims at improving water efficiency and competitiveness of the water industry; the recast Urban Wastewater Treatment Directive (UWWTD) and the Drinking Water Directive, stimulate energy neutrality and digitalisation of their sectors, pushing toward a water-energy nexus where hydropower can find room.

The report is organized into five main chapters. **Chapter 2** examines the state of the art and future developments of Hydropower, focusing on advancements in technology readiness, energy capacity, costs, and research funding. **Chapter 3** focuses on the value chain analysis, covering economic contributions, sustainability, and the role of EU companies in the market. **Chapter 4** provides an overview of the EU's global position and competitiveness in the Hydropower sector industry, analyzing market status, trade dynamics, and resource efficiency. **Chapter 5** concludes the report by synthesizing key findings and highlighting strategic opportunities and challenges

1.2 Methodology and Data Sources

The present report follows the general structure of all CETO technology reports and is divided into four sections with several indicators aiming to present and evaluate the EU xxxxx technology along its value chain:

- Technology State of the art and future developments and trends;
- Value chain analysis;
- EU position and global competitiveness.

The report uses the following information sources:

- Eurostat data;
- Existing studies and reviews published by the European Commission and international organisations;
- Information from EU-funded research projects;
- EU and international databases;
- EU trade data, trade reports, market research reports and others;
- JRC own review and data compilation;
- Stakeholders' input.

Details of specific sources can be found in the corresponding sections and Annex 2 provides a summary of the indicators for each topic, together with the main data sources.

2 Technology status and development trends

2.1 Technology overview

Technology description

Traditional hydropower is a renewable energy source that converts the hydraulic (water) power (potential and kinetic power) into mechanical power by means of a rotating turbine, and into electricity through the connection to an electric generator (Figure 2),

Hydropower plants can be of four main types (see the 2024 report for more details):

- Reservoir-storage hydropower plants (RSHP) (often called storage hydropower plants, or dam hydropower) are facilities that store water in reservoirs behind dams.
- Run-of-river (ROR) projects utilising the natural flow of water bodies and with limited storage capacity;
- Pumped-storage hydropower (PSH) is, beside RSHP, the main water and energy storage technology for power systems, with also pumping capacity. Closed-loop PSH stations (called pure PSH by Eurostat) are composed by reservoirs not permanently connected to natural watercourses and not permanently utilising natural (river) inflows, except rainfall and some groundwater inclusion. Open-loop PSH stations (also known as pump-back facilities and called mixed PSH by Eurostat) utilise also natural inflows from rivers, in addition to inflow from rainfall, creeks and groundwater. However, definitions vary depending on the country. Pumped storage hydropower represents more than 90% of global energy storage capacity, excluding RSHP¹.
- Hidden hydropower in water infrastructures: diversion schemes that utilize the available energy in conveyance systems for supply, transport and treatment of water and wastewater.

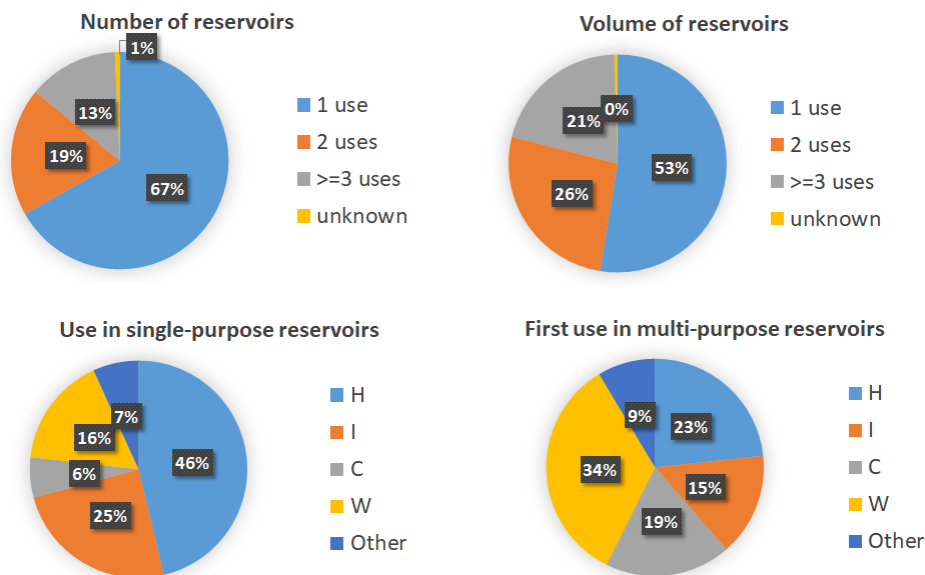
Traditional hydropower includes RSHP and ROR, while pure hydropower includes RSHP, ROR and energy generated in mixed PSH from natural inflow. In terms of size, hydropower stations are distinguished in large-scale and small-scale, with a typical threshold of 10 MW (installed power, variations exist). Hydropower can be classified, depending on the head, into high head, middle head, low head and very low head. Although a standard definition does not exist, it is reasonable to define a very low head when the head is below 5 m (2.5 m in certain cases ²), low head below 50 m, middle head between 50 m and 250 m, and high head above 250 m (see, e.g. Andaroodi and Schleiss, 2006 ³).

According to the International Commission on Large Dams (ICOLD), more than 59,000 large^b dams exist in the world, storing 8,919 Billion m³ at full capacity. In the EU there are 4,491 large dams according to the ICOLD 2023 register of dams and 67% are officially registered as single use (Figure 1). However, several dams, even if not officially registered as multi-purpose, provide more than one service and are, practically, multi-purpose. In the EU, 46% of single purpose reservoirs are for hydropower (which store 83% of the volume of single-purpose dams) and 23% of multiple purpose reservoirs have hydropower as a first use (34% have water supply as first use) (Figure 1). Furthermore, although not depicted in Figure 1, 6% of all reservoirs have hydropower as second use and 3.5% as third or fourth use. Overall, 48% of EU's reservoirs (associated to large dams) are with hydropower, i.e. the 46% of 67%, plus the 23% of the 33%, plus 6% and 3.5%. EU large-dam's reservoirs store 285 Billion m³ of water: 124 Billion m³ are in hydropower reservoirs as single use, while 111 Billion m³ are in multi-purpose reservoirs equipped with hydropower. It must be noted that not all the uses of a reservoir are always officially declared, and many hydropower reservoirs often provide additional services, like flood control, water supply for agriculture and for touristic purposes, not officially compensated or legally recognized.

^b higher than 15 m, or between 5 and 15 m and with storage capacity higher than 3 Mm³.

Figure 1. Top: Uses in large water reservoirs.

Below: use in single-purpose reservoirs (left, number of reservoirs), and first use in multi-purpose reservoirs (right): H (hydropower), I (irrigation), C (flood control), S (water supply), R (recreational), N (navigation), O (other use).



Source: ICOLD 2023

From the AMBER database⁴, at least 630,000 barriers were counted in European rivers (450,000 in the EU, and 225,000 if including only weirs and dams) and approx. 55,170 are defined as dams in the EU (the others include, e.g. ramps and other transversal barriers with no storage capacity). Around 25,000 hydropower plants exist in the EU (also including those in artificial canals, pipes and infrastructures). For further information on small hydropower in Europe refer to the UNIDO report⁵.

Sub-technologies

The hydropower sector is rather complex, as it uses electro-mechanical equipment (turbine, generator, motor generators and pumps for PSH), gearbox/transmission (mainly for small hydro applications), guide vanes or wicket gates to control the flow to the turbine, draft tube and spiral case or distribution pipes, the civil structures (e.g. weirs, the dam, tunnels, surge chambers, the powerhouse, penstocks, fish passages, spillways and canals), gates and valves, and the Control and Instrumentation equipment to monitor the status of the components (Figure 2). The power plant operation depends on the hydrological, environmental and market conditions.

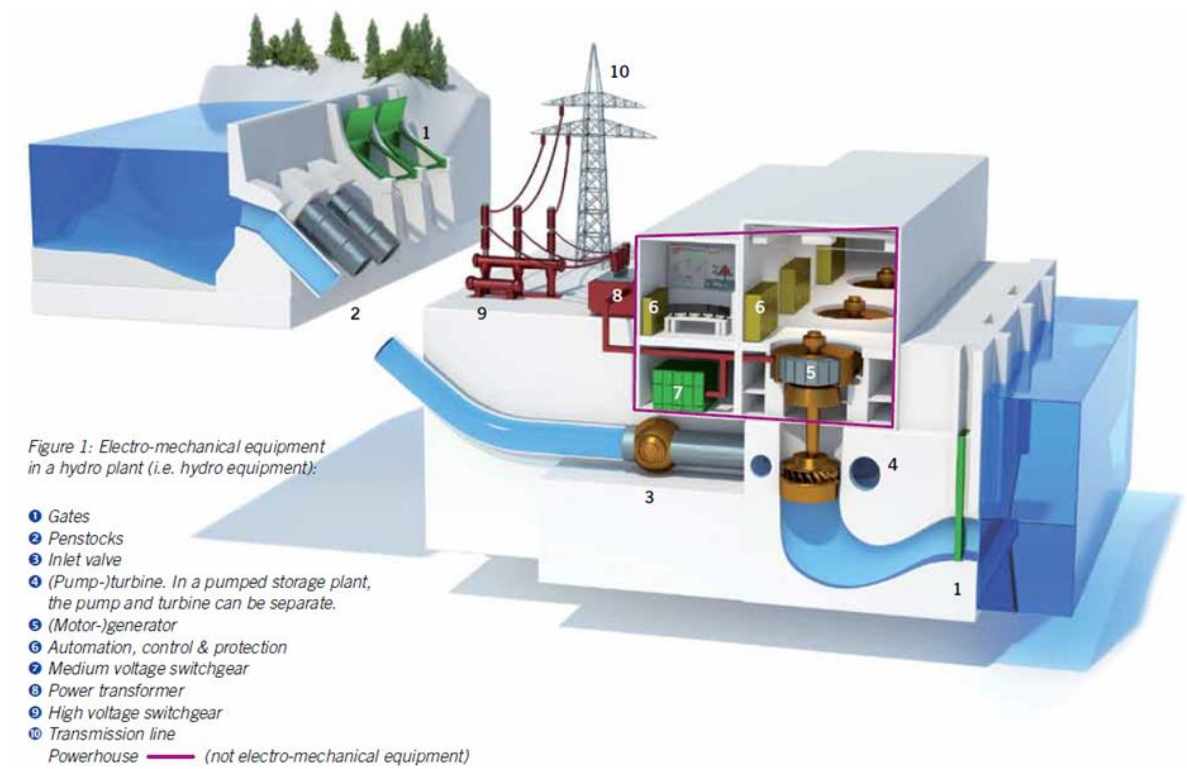
Hydropower turbines can be of three main types: action/impulse, reaction, gravity type. The overall efficiency of a high-middle head hydropower plant is generally > 80% and can exceed 90% at the optimal operating point, 5-times higher than photovoltaics and 3-times higher than wind turbines, while it generally range is between 60% and 80% for a very low head system.

With regards to the civil structures, the dam is one of the main elements, and it is a structure designed to retain water from a river. There are two types of dams:

- diversion dams, which use diversion systems to keep water surface at constant level and avoid changing the river regime. Diversion dams, also called weirs, are used for ROR, waterways, recreational activities;

- retention dams which create a barrier to store water in a reservoir, thus changing river regime and keeping water surface at a variable level. Retention dams can be built for two types of reservoirs:
 - a. supply reservoirs, in which water is extracted from the river for other uses, such as irrigation, navigation, drinking water, industrial use;
 - b. regulation reservoir, whose primary function is to regulate water flow. Water is stored and released into the river for several reasons, such as irrigation, flood protection, drought prevention, energy generation, compensation of irregular water releases of upstream power plants or other uses.

Figure 2. Sketch of a reservoir hydropower plant, with focus on the powerhouse.



Source: Hydropower Europe ⁶

2.2 Technology Readiness Level (TRL)

Nowadays, hydropower is an established sector and the largest and most flexible renewable electricity source, with well-known and robust technologies and construction methodologies. The most used turbines are the Pelton (high heads, low flows), Francis (middle and high heads, middle flows) and Kaplan-Bulb (low heads, high flows). Low-head Francis and Kaplan turbines can also be used as very low head converters. Reversible single-stage Francis-pump/turbines were developed in the early 1940s and spread in PSH installation with heads up to 700 m. A negligible share is represented by unconventional turbines, e.g. water wheels, Archimedes screws, Vortex turbines and Very Low Head (VLH) turbines, whose operating range is described in Quaranta et al (2021)². The installed capacity of Archimedes screws is typically below 100 kW and in 2012 around 400 installations were counted worldwide, 71 of them (2.5 MW) in Europe (Lashofer et al., 2012⁷). VLH turbines typically work below 500 kW and their total installed capacity in Europe is around 30 MW

(MJ2 Technologies, 2023). Water wheels are generally below 30 kW and with average power of 13 kW in the EU (Quaranta et al., 2022b²⁸). In the EU, 12 Vortex turbine installations were counted, with a cumulative power of 78.1 kW (in Austria, Germany and Belgium)⁸, but their number is estimated to higher as not all turbines are counted in that paper. In the EU + UK, according to Voith Hydro database, 24 Banki turbines are installed with a total power of 10.4 MW and maximum head 170 m (updated to 2020), but that are much more according to the company Ossberger which manufactures them.

Nevertheless, R&D activities are continuously ongoing and novel technologies and operation methods are under investigation. Being hydropower systems very complex, most of challenges nowadays are on how to integrate energy, environmental and equipment issues, finding novel management and operating ways. For example, the operation of reservoirs can be improved by better real time control, implementing novel methods for environmental control and management. Efforts are needed to improve the operation of hybrid systems, like when hydropower is integrated with batteries, hydrogen and photovoltaics. All these technologies mainly combine sub-technologies coming from different applications and it is difficult to really assign a TRL. Fry et al., (2022)⁹, in the EU funded project ETIP Hydropower, estimated that between EUR 190 and 324 Million are required to be invested by funding schemes in the following topics: flexibility, optimization of operation and maintenance, resilience of electro-mechanical equipment, resilience of infrastructures and operation, environmentally compatible solutions and mitigation of global warming impacts.

When analysing the electro-mechanical equipment, most of novel technologies are already at their first stage of commercialisation and a few installations already exist, thus their TRL is already high. Kougias et al., (2019)¹⁰ and Oladosu et al., (2021)¹¹ reviewed these emerging hydropower technologies. Quaranta et al., (2021)¹² assessed how new innovations and novel strategies could improve hydropower operation (e.g. digital tools, new turbine designs). The innovative materials are discussed in Quaranta and Davies (2021)⁶³: some materials have not been systematically implemented, and more research is needed. Environmentally enhanced turbines are discussed in Quaranta et al. (2021)¹³ and are already commercialized, although only a few cases exist, and more research is needed to better optimize them. In order to respond to the increasing needs for flexibility^c of operation, hydropower systems need to reach higher levels of digitalisation. Digital solution can also be implemented to mitigate environmental impacts⁷⁵; several technologies and new sensors are under development and are implemented also for environmental purposes (e.g., sensors for fish, sediment management).

Looking at specific technologies, some examples are the following ones. At some emerging low head hydropower turbines TRL = 8 can be assigned, as the actual systems are manufactured by a few companies and are not yet fully deployed (Quaranta, 2026). More R&D and investments are needed to make them fully commercializable, reducing costs and improving their reliability and performance knowledge under varying operative conditions. The Hydrostatic Pressure Machine and the Counter-rotating rotor turbines have a TRL=4-5. Very small hydropower turbines, like the ones developed in the EU project H-HOPE have a TRL around 4. Some turbine innovations, like splitter blades or the X-blade turbines are not fully deployed and need some research, although they have already been implemented in a few cases (TRL = 8)¹². The draft tube behaviour can be optimized by permanent passive techniques (with no external power source) and active adjustable techniques (with external power source) and they are at TRL 4-5¹⁰. The short-circuit operation, and the variable speed operation are already in place, but deeper investigations are needed to unlock their potential. New vibration adsorbers, bearing and seal systems are continuously developed. New configurations for established turbines are innovations applied in specific cases, thus with a certain degree of innovation although they use proven technologies (e.g., the case of Stegenwald hydropower plant in Austria^d, among others).

c The ability of a system to respond to changes in generation and/or load is called system flexibility.

d <https://www.global-hydro.eu/en/blog/world-first-turbine-innovation-benchmark-in-economic-efficiency>

2.3 Installed energy capacity and energy generation

In 2024, the global installed power of grid-connected hydropower reached 1443 GW, including 189 GW of pumped-storage hydropower (PSH), with an annual generation of 4587 TWh, 10% more than in 2023^{14,15}. Hydropower also provides 1.8 GW of off-grid hydropower electrification services, mainly in Africa, South America and Asia¹⁶.

According to Xu et al. (2015)¹⁷, Europe as a continent, has a hydropower technical potential (thus considering the potential that can be exploited with the existing technology, independently from economic factors) of 1,121 TWh per year (hereinafter, TWh/y), and, with a current production of 569 TWh/y and installed capacity of 258 GW, has developed approx. a half of the available technical potential, the highest share, globally. There is still untapped technical sustainable potential in Europe (more details in section *Future projections*).

In this chapter, installed power and annual energy generation are discussed, considering both traditional hydropower and PSH. It must be noted that, while traditional hydropower is classified as renewable energy technology, PSH is classified as energy storage technology. However, mixed PSH, where there is freshwater inflow and, generally, the annual energy generation is higher than the energy used for pumping, can also be classified as renewable energy technology when considering the net energy generation (energy generation from water not previously pumped, data available from EUROSTAT). By traditional hydropower we refer to reservoir and RoR hydropower with no pumping capacity. By pure hydropower we refer to the previous one plus energy generation from mixed PSH from water not previously pumped.

Installed power

According to Eurostat, in 2023, the hydropower installed capacity was 153.2 GW in the EU, mostly located in alpine environments. Comparing to 2015, this corresponds to +6 GW of installed capacity, and 12 GW are estimated to have been refurbished (combined Eurostat data and Production data from this CETO report). Out of the total installed capacity, 106.6 GW comprises “traditional” hydropower stations, meaning hydroelectric facilities that solely serve electricity generation (including multi-purpose services) (24.6 GW is RoR). Another 22.7 GW refers to closed-loop pumped-storage hydropower. The remaining 23.9 GW is mixed hydropower, meaning typical PSH facilities connected to freshwater bodies, generally rivers^{18,19}. In Table 2 the EU’s hydropower fleet is classified by type and installed power, updated to 2022^e. A review of small hydropower in the EU was published by Quaranta (2026)²⁰.

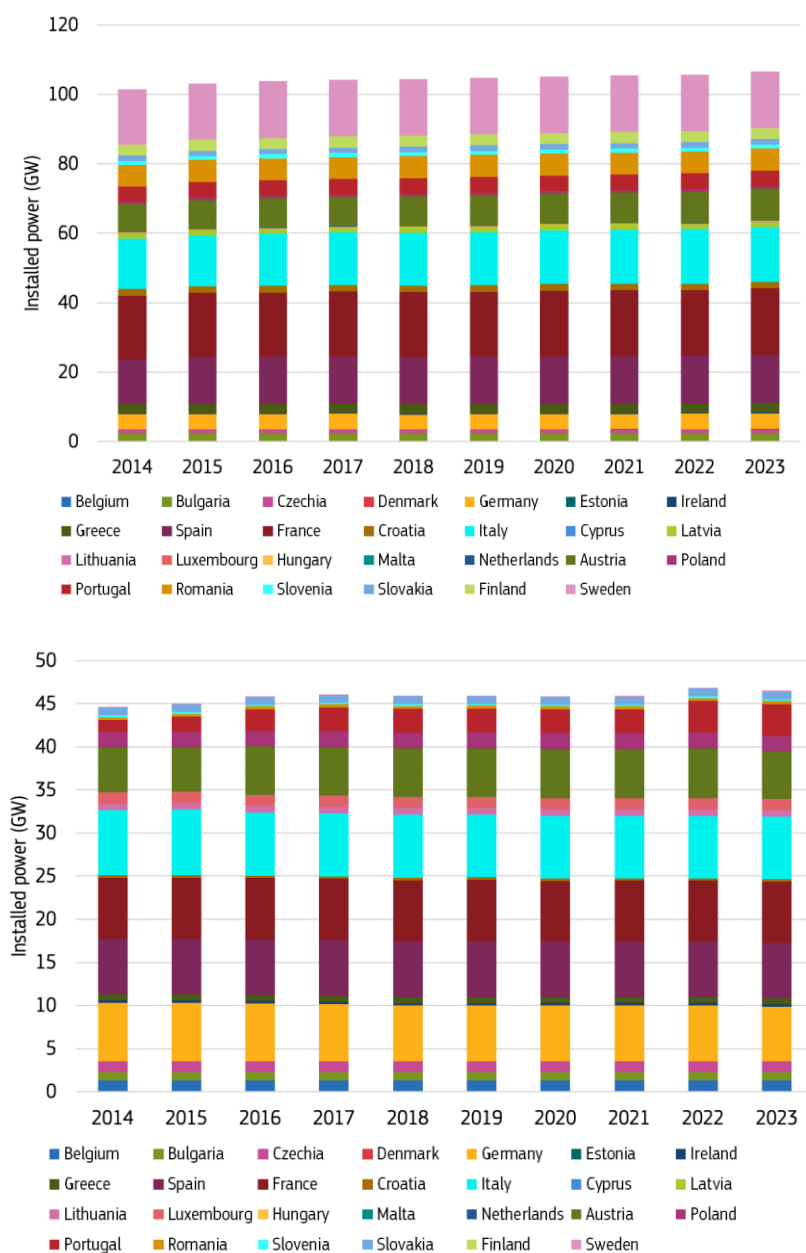
Table 2. Installed turbine power (in % on the total of 152 GW) at the EU level, categorized depending on hydropower plant type (RoR, RSHP, PSH) and installed power class P (MW). For PSH the turbine’s installed capacity is considered. 8 PSH are below 10 MW with overall turbine installed capacity of 44 MW.

P (MW)	Share (%) on total Power			
	RoR	RSHP	PSH	Total
$P > 10$	16.8%	44.5%	28.4%	89.7%
$1 < P \leq 10$	4.6%	2.8%	0.0%	7.5%
$P \leq 1$	2.1%	0.0%	0.0%	2.1%
Total	23.5%	47.4%	28.4%	100.0%

Source: combination of data of the JRC hydropower database, Voith Database and UNIDO database (year 2022).

^e The total installed capacity and the total installed capacity of ROR are higher (+3 GW, +15 GW) than that from other sources (e.g. Eurostat) since it is difficult to track exactly the hydropower fleet composition, and, probably, some hydropower plants with small pondages are considered RoR in the JRC database.

Figure 3. Installed power for traditional hydropower (up) and PSH (bottom figure), generation capacity (GW).

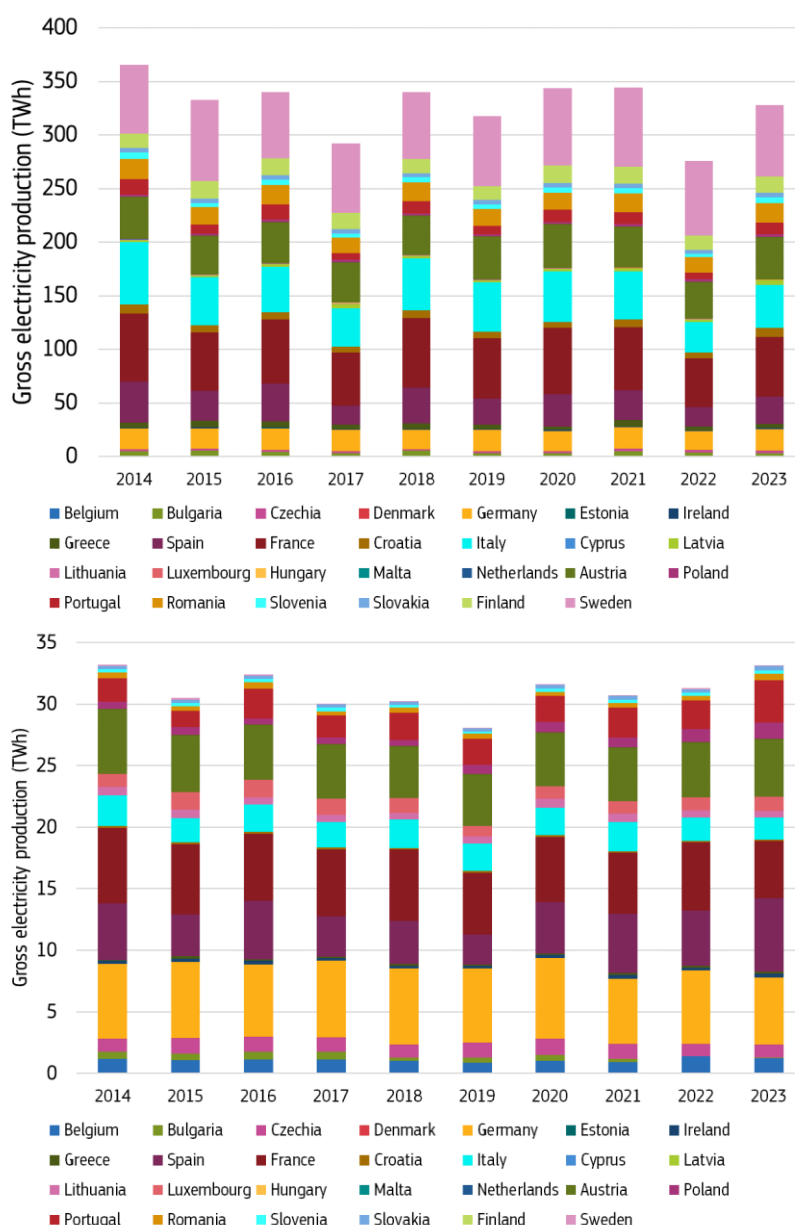


Source: Eurostat, 2025.

Annual energy generation

In the last decade, the annual energy generation from traditional hydropower and PSH (excluding PSH) in the EU has oscillated between 272 and 365 TWh/y depending on the hydrological conditions with the average value being 328 TWh/y. The lowest was in 2022, mainly due to the water scarcity in several regions. Despite this, the total installed capacity increased, and the operation of closed-loop PSH increased by 15%, both in generation and in pumping mode. Energy generation from PSH (closed-loop PSH + energy from water previously pumped in open-loop PSH) ranged between 28 and 33 TWh, with an average of 31.3 TWh. In this document, energy generation of mixed PSH was split into two parts: the one from water previously pumped, and included in the category pure PSH, and the one from natural inflow, included into pure hydropower.

Figure 4. Annual electricity generation for pure hydropower (up) and pure PSH (bottom figure) (TWh).



Source: Eurostat, 2025.

Current Energy Storage

Globally speaking, energy storage in reservoir hydropower^f RSHP and PSH is the largest source of energy storage today and is the only mature technology for long duration energy storage (IEA, 2023b²¹, World Bank, 2024²²). Globally, reservoir hydropower has a storage capacity of about 1500 TWh, which is about 170 times more than the global fleet of PSH and almost 5000 times more than the Li-Ion battery capacity of stationary applications (300 GWh in 2023), and more than 2000

^f reservoir hydropower= a particular type of hydropower system that uses a reservoir and that is not provided with pumping capacity;

hydropower reservoir: term that specifically refers to a reservoir with single use hydropower

times when also considering automotive applications (IEA 2021²³). Excluding reservoir hydropower, more than 90% of the remaining energy storage in the world is in PSH^{24, 25}.

Quaranta et al., (2024) estimated that the available theoretical potential of energy storage is 55 TWh^g in the EU's hydropower reservoirs (excluding PSH and neglecting the cascade effect^h of reservoirs, that, for example, in Sweden increases energy storage from 8 TWh to 30 TWh). The theoretical energy storage in PSH is 6.3 TWh. Results reported in Graabak et al., (2017)²⁶ show that in the EU the energy stored in water reservoirs is approx. 90 TWh, which is higher than the one estimated above of $55+6.3=61$ TWh, and this is probably due to the cascade effect considered in the Swedish estimate. Considering some reduction factors, the technical usable storage capacity of PSH reservoirs was finally estimated at around 500-800 GWh from Quaranta et al. (2024) for PSH, and in 25 TWh for RSHP (without cascade effect). 156 PSH stations operate in the EU with an overall turbine capacity of 46 GW (the installed pumping capacity is 39 GW in the EU), and 9 GW are under construction or planned. Several PSH projects were either announced or achieved noticeable progress in Europe during 2024. PSH still has remarkable untapped potential, as shown in Table 3. Grid battery storage was 61 GWh in the EU in 2024.

When considering Europe as a whole, it is estimated that European hydropower reservoirs provide a theoretical storage capacity between 180 TWh³⁵ and 220 TWh²⁷. Norway has the largest (85 TWh) storage capacity in Europe, more than any EU Member State. Switzerland hosts 8.4 TWh. Norwegian hydropower acts like a battery for balancing variable renewables in neighbouring countries, with more than 1000 reservoirs (Graabak et al., 2017)²⁶.

Capacity factors

Hydropower productivity is not uniform across the EU and reflects the climatology, topography and water resources of each region and the power plant type prevalence. This variability is typically shown by the Capacity Factor (CF) that is expressed as the ratio of annual electricity generation to the electricity that would be generated if the power plant would work the whole year at nominal power. The average CF in EU during 2014-2023 was 29% for RSHP and 37% for RoR, which is higher than CF of the photovoltaics and wind sectors.

Scientific assessments of available potential

The modernization of the existing hydropower fleet is an attractive opportunity to increase efficiency, production, flexibility, sustainability and resilience to climate changes (the average age of the EU' hydropower fleet is almost 45 years. Quaranta et al., (2021) estimated that the annual electricity generation from the existing hydropower EU's fleet could be increased by approximately 10% (~40 TWh/y)ⁱ by increasing the efficiency and with no increase of installed original capacity, implementing new tools for hydropower digitalisation, modern electro-mechanical equipment (to replace deteriorated equipment, increase its flexibility and slightly the best efficiency point), and by optimizing the waterways (e.g., reducing friction and head losses). Additional strategies to increase generation from the modernisation of aged plants include dam heightening (useful especially to increase storage capacity), new waterways to increase the peak installed capacity and a better reservoir management to reduce spills (the latter, with possible increase of annual revenue up to 10%). Modernization should also aim at improving resilience to climate changes and to optimize

^g calculated by $head \times stored\ volume \times coef.$, thus assuming: $head \gg$ dam height and constant at its maximum value while the reservoir empties, and the full design volume ($coef.$ is the conversion factor to obtain TWh). The larger reservoir is used for the calculations, since we are referring to the maximum theoretical one and because from ICOLD only the larger reservoir is generally traceable.

^h defined as an operation where downstream reservoirs store and release water coming from upstream reservoirs, in a combined, integrated and optimized system

ⁱ Under the assumption of the market and hydrological conditions at the time of the study.

hydropower operation to better deal with future energy needs, market dynamics and new hydrological conditions (more floods and droughts).

Developing hydropower in existing infrastructures (generally, small hydropower, even micro hydropower), e.g. in water distribution networks (aqueducts), in existing low head barriers (e.g. water wheels in water mills) and in wastewater treatment plants exhibit an untapped potential of 10 TWh/y in the EU ^{28, 29}. Small-scale hydropower opportunities integrated in existing facilities can provide decentralized energy when the electric grid is not available, difficult to be connected or to avoid further expansion of the grid (e.g., some alpine areas, mountain refugees). There also exists a hidden potential in hydraulic infrastructures in the private water intensive industry, such as mining or energy production (using cooling waters). Hydrokinetic turbines exhibit a low potential, but could represent an interesting strategy when installed at the hydropower plant tailrace as kinetic energy dissipation devices (Quaranta and Muntean, 2023)³⁰. New small hydropower plants that involve the construction of new weirs/dams are rather controversial, due to their impacts and limited annual energy generation with respect to the existing hydropower fleet. Bodis et al. (2015)³¹ estimated that the potential in the EU is 111 TWh/y, which is approx. 40 TWh/y more than the current one.

Pumped Storage Hydropower

For PSH, some projections foresee an increase in PSH capacity up to 70-75 GW by 2050 ^{32, 33}. POTEnCIA (Annex 5) assumes an increasing use of PSH of +67 TWh and 8.3 GW in 2050 with respect to 2025, and most of the power capacity of new storage technologies is assumed to be mainly from batteries. The results of Stocks et al. (2021) and Hunt et al. (2020), better contextualized for the EU in Quaranta et al. (2024)³⁴, showed a quite high remaining potential for PSH, although the installation costs could be higher due to the fact that they would be built in the most expensive sites (the most suitable are already exploited) and also because of environmental mitigation measures to be implemented. Gimeno-Gutiérrez and Lacal-Aránzategui (2015) and Weber et al. (2024) assessed the potential of interconnecting reservoirs and exploiting abandoned mines, respectively, whose Capital Expenditure (CAPEX) and environmental impacts would be noticeably lower than those of new greenfield projects. The estimated economic potential that could be realized would be 0.8 TWh and 3 TWh (considering the most economic sites), which is more than 50% of the current theoretical one of PSH. Reservoir interconnection, abandoned mines, virtual energy storage gains resulting from the spatio-temporal coordination of hydropower across Europe (especially the connection with the Norwegian grid) and the upgrading of existing PSH (including dam heightening) are opportunities for pumped-storage hydropower and to balance generation variability^{35,36}. Underwater PSH and low-head energy storage³⁷ are also under investigation, and the associated potential of the third one (i.e. abandoned mines) is approx. 5 TWh (3 TWh the least expensive sites). Energy could be stored using existing lakes, small depressions or retention basins on a terrain in sustainable urban drainage systems. In France, the potential of PSH from small lakes and reservoirs was estimated to be about 33 GWh, which, according to Quaranta et al. (2024), is approximately 38% of the current national technical energy storage capacity of PSH ³⁸. Existing reservoirs can be paired to new reservoirs at higher altitudes. In Switzerland, dam heightening implemented at the national scale would increase storage capacity by 30%³⁹.

A real technical opportunity for PSH expansion is first of all expanding the operating range of the existing ones, e.g. by introducing smart sensors, variable speed turbines with increased efficiency and system optimization. Countries with redundant PSH potential could look for increasing their potential market share by selling their services to neighboring countries. Opening cross-border markets for balancing capacities could also serve as an important incentive to increase the use of the existing PSH capacity¹⁸.

Additional strategies to increase generation include hybridization with other energy technologies. For example, run-of-river plants can produce hydrogen when energy prices fall to zero. Batteries ⁴⁰ can ensure energy storage for several hours, whilst hydropower can store and supply energy for

days and weeks, including seasonal transfer, and batteries and hydropower can be integrated together⁴¹. Reservoirs can host floating PV, or PV could be installed on dam surfaces⁴². Methane capture processes are under investigation, but R&D is further needed to improve cost-effectiveness (this measure is especially relevant for tropical reservoirs, while not very relevant in European reservoirs, with a few local exceptions)³⁰.

Table 3 summarizes the hidden potential in the EU (or Europe) assessed in scientific studies^j. For further details refer to Perpiña Castillo et al. (2024).

Climate changes

Hydropower generation depends on water availability and may suffer from water shortages in long dry periods. On the other hand, optimal management of hydropower reservoirs, along with a better inflow and weather forecast, can help in mitigating droughts and can act as a flood control system. Gøtske and Victoria (2021)⁴³ estimated that the annual inflow for high (mid)-emission scenarios is going to decrease by 31% (20%) in Southern countries and to increase by 21% (14%) in Northern countries (especially in winter periods), and more frequent and prolonged droughts in Mediterranean countries are expected. The median decrease of generation of ROR hydropower will be -3% in the future on the Italian Alps. For further details see the 2022 CETO report. Farinotti et al., (2016)⁴⁷ suggested that the deglaciating catchments (catchments where glaciers are disappearing) could make an important contribution in increasing the production in the future (by 2050) if the existing hydropower fleet in such catchments were upgraded by increasing the storage volume (for RSHP) or production capacity (for ROR and RSHP). New development should also aim at increasing storage capacity of existing reservoirs, by de-sedimentation; the average annual sediment inflow into European reservoir is 0.7% of the reservoir volume. Prevention and mitigation solutions (e.g. bypasses)⁴⁴ can help in mitigating this issue; dredging is also an option, but costs several billions EUR per year in the EU⁴⁵, and generates adverse impacts on surface waters (both changes in morphological characteristics and increased chemical pressure because of possible contaminated sediments).

PSH could contribute to adaptation to climate change, by providing valuable long-term water storage, storing water during flood events, and using the water during drought periods. Large PSH can store energy on a seasonal scale, storing excess solar power during summer to generate electricity during winter⁴⁶. In alpine countries, new reservoirs could mitigate the already visible effects of climate change (floods, droughts and fire, and to compensate hydrological changes due to glacier retreat⁴⁷).

^j Some of these studies have used the European hydropower database developed by the JRC, including coordinates, type, installed power, and, in most cases, gross head, annual energy generation and water storage capacity.

Table 3. Potential of different hydropower and PSH strategies (in the EU when not specified). For PSH, results represent energy storage potential, while annual energy generation for the others.

Strategy	Potential	Reference and Comment
Closed-loop pumped-storage hydropower (PSH)	339 TWh (25 TWh the cheapest sites)	Stocks et al., (2021) ⁴⁸ , EU, theoretical energy storage
Open-loop (or, mixed) seasonal PSH	50 TWh (10 TWh the cheapest sites), or 173 TWh with cascade effect	Hunt et al., (2020) ⁴⁹ , EU, theoretical energy storage
Spatio-temporal coordination of hydropower reservoirs	140 TWh	Wörman et al., (2022) ³⁵ , interconnection between 1200 km and 3000 km, Europe, reduction of energy storage demand over 3-5 years
Reservoir interconnection	29 TWh in Europe, 4 TWh in the EU (max 20 km distance among reservoirs), 198 GWh and 141 GWh, respectively (max 5 km distance).	Gimeno-Gutierrez and Lacal-Arantegu (2015) ³⁶ , theoretical energy storage
Offshore sea water PSH	156.5 TWh	Kougias et al., (2019) ¹⁰ , Puchta et al., (2017) ⁵⁰
PSH in mines, underground PSH and low-head PSH	5 TWh of PSH in mines	Menendez et al., (2017) ⁵¹ , Hoffstaedt et al. (2022), Weber et al. (2024) ⁵² , more details in Quaranta et al. (2024) ³⁴
Hydropower plant modernization	40 TWh/y	Quaranta et al., (2021), Quaranta and Muntean (2023), additional energy generation
New reservoir-hydropower in deglaciarized areas	11 (0.76) TWh/y	Farinotti et al. (2019) ⁵³ , energy generation from all (and from the most suitable) sites
Powering of Non-Powered Dams (NPDs)	1.75 GW	Garrett et al., (2021) ⁵⁴ , Europe
New RoR small hydropower plants	30-70 TWh	tens of TWh/y, Quaranta et al., (2022), and Bodis et al., (2015), summarized in Quaranta (2026), but would entail new barriers in rivers.
Existing historic barriers not mill-related	5.2 TWh/y	EU+UK, Punys et al., (2019)
Water Utilities (Pressurized WDNs and WWTPs)	3.0+0.1 TWh/y	EU+UK, Quaranta et al., (2022), 0.75 TWh instead of 3 TWh if considering new estimates and capacity factor 50%
Heat recovery from generators	2.9 TWh/y	Quaranta and Muntean (2023)
Water wheels in existing mills	1.6 TWh/y (3 TWh/y using all types of turbines)	Quaranta et al., (2022) ²⁸
Rainfall on building roofs	0.5 TWh/y	Quaranta et al., (2022b) ⁵⁵
Hydrokinetic turbines in rivers	0.17-1.2 TWh/y	Quaranta et al., (2022) ²⁸
Pressurized conduits for irrigation and industrial flows	<0.1 TWh/y	EU+UK, Mitrovic et al., (2021) ⁵⁶
Floating PV (evaporation reduction)	<0.1 TWh/y	Quaranta et al., (2021), 10% of reservoir surface coverage
Floating PV (FPV) on hydropower reservoirs	139 TWh/y/729 GWp	Kakoulaki et al., (2022) ⁵⁷ , covering 10% of 1608 km ² of EU's reservoir surface, associated to 49 GW of hydropower installed capacity and 94 TWh/y of hydropower generation / Lee et al., (2020) ⁵⁸ , Europe, 14% of reservoir's surface coverage

2.4 Technology Cost

Hydropower is financially competitive with other electricity technologies, achieving some of the lowest values of electricity generation costs (LCOE). IRENA and the World Bank's analysis identified hydropower as currently one of the most economical forms (in the long term, considering the life span) of renewable electricity generation ⁵⁹. One of the main advantages of hydropower stations is that the operating cost is low and generally very stable since it does not depend on fuel cost. Moreover, hydropower stations have a long service life (assumed 40–50 years), with the civil works even reaching 80–100 years. However, new large hydropower projects are capital intensive, requiring large upfront investments. Licensing and construction periods can be long and complicated especially in large-scale projects (several years and in certain cases even exceeding 10 years).

Table 4 depicts the approximated shares (in %) of civil, mechanical and electrical components in the total capital cost of different hydropower plants.

Table 4. Capital cost breakdowns for the main types of hydropower in Europe, 2021. Civil: Dam, tunnels, piping, powerhouse (aboveground or underground), roads. Mechanical: turbine, penstock, gates, valves, hydraulics. Electric: generator, transformer, cabling, grid connection.

Hydropower type	Civil	Mechanical	Electric
Large-scale Reservoir hydropower	70%	10%	20%
Run of river (Large-scale and small-scale)	50%	30%	20%
Pumped-storage hydropower	50%	30%	20%

Source: IRENA, 2025.

LCOE

In 2024, the global weighted-average (based on the power) LCOE for new hydropower stations (greenfield projects) was USD 57/MWh, 30% higher than in 2010, but lower than the 58 USD/MWh of 2023 ⁵⁹. In 2024, the global weighted average total installed costs of newly-commissioned hydro projects decreased 21%, year-on-year, to USD 2,267/kW. However, the economic benefits associated to the multiple services provided by hydropower are not included in the LCOE (flexibility, ancillary services, water management and supply of reservoirs, among others). 96% of the hydropower projects commissioned in 2021 had an LCOE lower than the range of newly commissioned fossil-fuel fired capacity cost. Moreover, 85% of the hydropower capacity commissioned in 2021 had a LCOE lower than the cheapest new fossil fuel-fired cost option ⁵⁹.

The LCOE cost in Europe are depicted in Figure 5. The LCOE of new large hydropower projects (covering all types) in Europe including Turkey is approx. 69 EUR/MWh with a range between 22 EUR/MWh (5° percentile) and 261 EUR/MWh (95° percentile); for small hydropower, the weighted average is 89 EUR/MWh with a range between 82 EUR/MWh (5° percentile) and 234 EUR/MWh (95° percentile) ⁵⁹. Large run-of river projects may cost, on average, 40 EUR/MWh, whereas reservoir hydropower, including pumped-storage power plants, around 100 EUR/MWh. The generation costs of small hydropower projects (below 10 MW), generally run-of-river, typically are 40% to 60% higher. When considering refurbishment projects, the cost ranges between 20 and 30 EUR/MWh⁶.

CAPEX

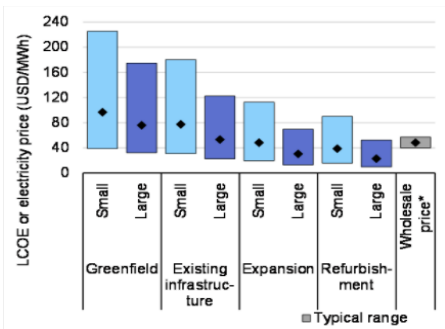
The CAPEX of recent large hydropower projects of all types is USD 2,200 per installed kW (including civil works) with a range between 757 USD/kW (5° percentile) and 8,994 USD/kW (95° percentile). Run-of-river power plant projects may be situated in the range of 1,000 EUR/kW to 1,800 EUR/kW with average at 1,400 EUR/kW, whereas reservoir hydropower including pumped-storage hydropower are in the range of 1,400 EUR/kW to 4,000 EUR/kW with an average of about 2,500 EUR/kW⁶. The CAPEX of small hydropower projects (below 10 MW), generally run-of-river, typically are 40% to 60% higher, and ranges between 1,434 USD/kW and 10,356 USD/MW with weighted average 5080 USD/kW.

Micro-hydropower projects generally cost up to EUR 5000/kW (e.g. water wheels in old mills, hydrokinetic turbines, turbines in pressurized water networks and low head turbines in existing barriers or weirs). The cost of projects that use parts of water management infrastructure (e.g. existing reservoirs, non-powered dams and conveyance systems) and brownfield projects that expand operational power plants or replace their equipment (a main future trend in Europe), is typically up to 70% lower than for new ones, as spending goes mainly towards replacing or adding electro-mechanical equipment. The cost per kW of run-of-river power plants is investigated in detail in Patro et al. (2022)^{k, 60}.

O&M

Annual operation and maintenance (O&M) costs are often quoted as a percentage of the investment cost per kW per year, with typical values ranging from 1% to 4%. Operation and salaries take the largest slices of the O&M budget. Operational cost varies from 20% to 61% of total O&M costs, while salaries vary from 13% to 74%, and materials between 3% and 4%⁵⁹. The International Energy Agency (IEA) assumes O&M costs of 2.2% for large hydropower projects and from 2.2% to 3% for smaller projects, with a global average of around 2.5% (IEA, 2021⁵⁹). This would put large-scale hydropower plants in a similar range of O&M costs – expressed as a percentage of total installed costs – as those for wind, although not as low as the O&M costs of photovoltaics. When a series of plants are installed along a river, centralised control, remote management and a dedicated operation team to manage the chain of stations can reduce O&M costs to much lower levels and increase generation^l.

Figure 5. LCOE costs and wholesale price of hydropower in the European Union⁶.



Source: ETIP Hydropower

^k In Patro et al., (2022) equations to estimate the cost of different ROR plants (diversion, in-stream, canal type) are presented. They showed that the highest LCOE is for diversion type, that is 20% higher than the LCOE of dam-toe and canal type hydropower plants, but the investment cost mainly depends on the installed power rather than on the plant type.

^l In Europe, this is not really an issue since there are very few cascade hydropower schemes which do not belong to the same operator. A cascade hydropower system owned by the same operator is today exploited systematically in a coordinated way in order to maximize generation and benefits in view of market demand.

Projections

Hydropower plants are site-specific and each project has unique design characteristics, and a clear projection of costs for the next decades is hard to find. Accordingly, in regions where the best locations have already been exploited (e.g. in the EU), the remaining technical potential usually refers to less advantageous sites with higher installation costs. Europe and North America have the highest hydropower investment costs because of relatively high labour costs, fewer undeveloped economical sites and steep fees to mitigate impacts on the environment and on existing infrastructures. Therefore, installation costs of greenfield hydropower projects may increase in the future. Furthermore, a big share of the installation cost of a hydropower project is for the civil works, whose cost typically increases at rates subject to construction cost inflation⁶¹. Therefore, the evolution of generation and installation costs up to 2050 is mostly influenced by the market inflation.

Due to improvement in construction efficiency, mainly of underground works, construction costs may be reduced for reservoir and pumped-storage hydropower power plants. Nevertheless, this potential saving will be compensated by the higher costs of environmental mitigation measures, and the expected increase of the investment costs could be better faced if adequate remuneration will be paid for the multiple services and benefits. For example, pumping operation of PSH helps to balance excess supply and avoid curtailment; however, grid operators do not currently remunerate this service apart from lower prices, and this service will become increasingly important to avoid renewable curtailment. Current power market conditions show that the revenues attained from the price differential (arbitrage) along with the declining revenues from the provision of the ancillary services are not enough to cover the fixed investment and administrative costs of these units. Several sites have been placed on indefinite hold due to a perceived lack of profitability, including Lago Bianco 1000 MW and Grimsel III 600 MW in Switzerland, Atdorf 1400 MW and Waldeck II plus 300 MW in Germany (IHA, 2022, Voith Hydro, 2023, pers. comm.).

Hydropower projects have longer pre-development, construction and operational timelines than other renewable energy technologies, hence investment risks for large hydropower schemes are higher, requiring specific policy instruments and incentives, as well as a longer-term policy perspective and vision. However, renewable energy policy attention in the past two decades has focused primarily on wind and solar PV technology expansion (and lowering their cost), mainly through support schemes such as deployment targets, financial incentives and long-term power purchase contracts. As of 2022, more than 100 countries have introduced short- and long-term targets and financial incentives for wind and solar PV, but fewer than 30 have policies targeting new and existing hydropower plants⁶².

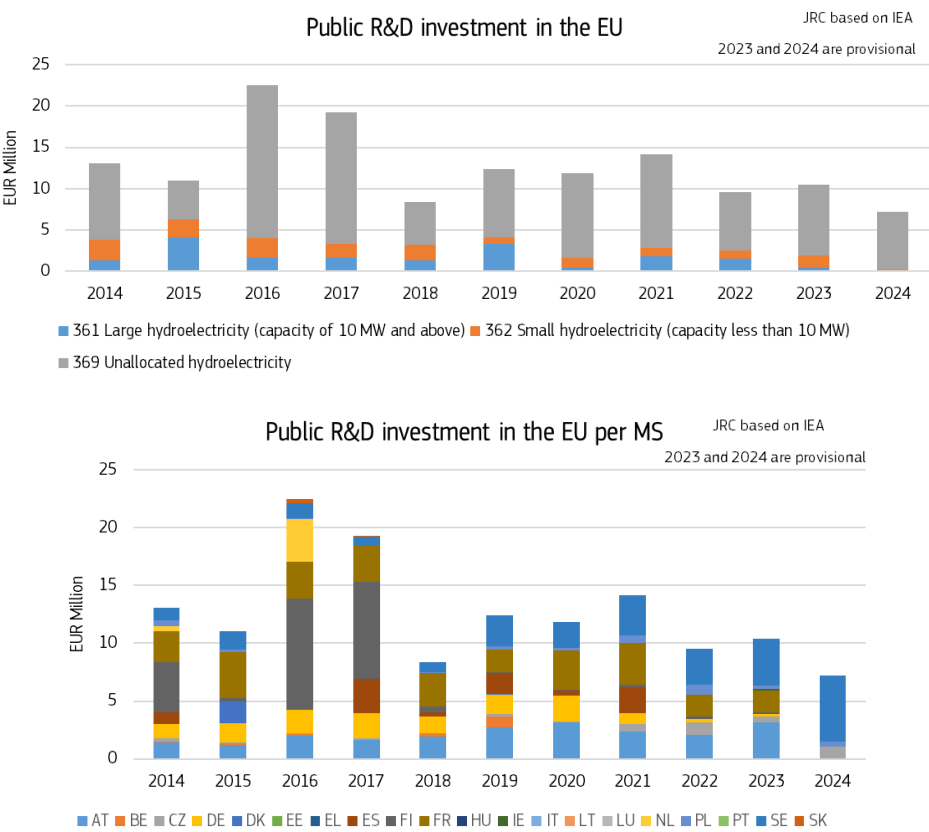
2.5 Public RD&I funding

Despite hydropower technological maturity, research efforts are still ongoing to develop new concepts and technologies¹⁰, novel materials⁶³ and innovative projects⁶⁴. Recent hydropower research and development (R&D) efforts mainly aim at improving the performance of systems and components, sustainability and readiness of hydropower for modern power markets. Hydraulic design and mechanical equipment R&D focus on expanding the flexibility of stations to support a wider range of operation, increasing efficiency levels, and to minimize the environmental impacts, like sediment transport, fish migration impairment and hydropeaking. In the last decade (2014–2023), public spending for R&D on hydropower in the EU ranged between EUR 8.4 Million and EUR 22.5 Million per year (Figure 6) (2024 data not complete yet), with decreasing trends after peaks. The main hubs of public spending are Austria, Germany, Finland, France and Sweden. Annual public spending in hydropower R&D is generally not stable as it follows the implementation of targeted actions, short-term national policies and specific EU calls.

Figure 7 depicts the annual R&D public spending in EU Member States (MS). Furthermore, while in certain MS funding is somewhat stable (Germany, France, Sweden), in several others it is irregular

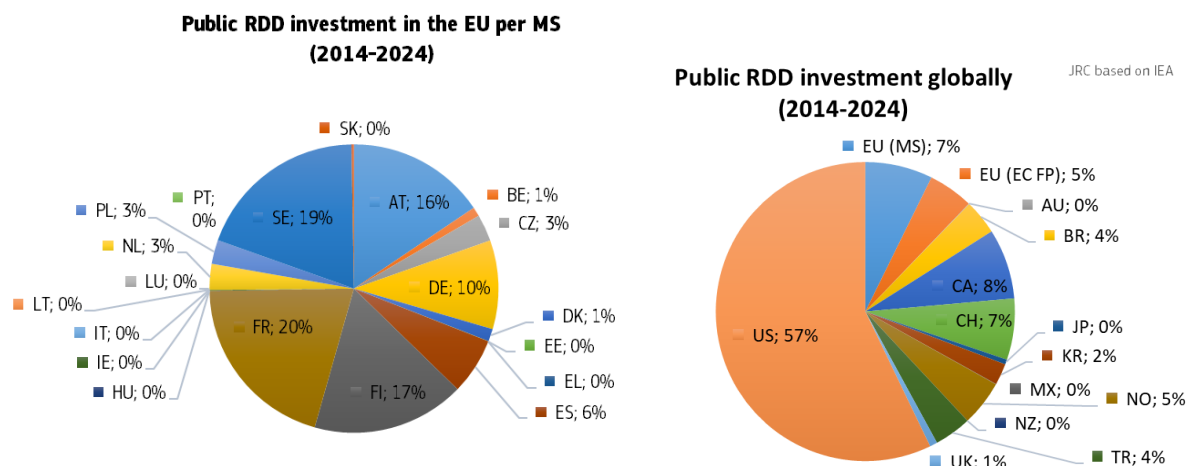
and dominated by targeted investments in specific years. Hydropower public spending was nearly 24 times lower than that for wind and 17 times lower than that for solar PV in 2024.

Figure 6. EU public R&D investments [EUR Million]. Source: JRC based on IEA data. Unallocated=not specified.



Source: JRC

Figure 7. Public R&D investments in hydropower for the EU member states over the period 2014-2024.
Source: CIndECS2022. For the non-EU acronyms, see the Nomenclature list. EU (EC FP) indicates funding form EU framework programmes (H2020) and is only available from 2014 onwards.



Source: JRC

The average public R&D investment is EUR 13.4 Million on annual basis in the EU (2013-2023), slightly higher than the annual public spending in Canada (approximately EUR 12.7 Million annually) and higher than that of Norway (about EUR 10 Million), and similar to Switzerland (about EUR 13.95 Million). In the U.S., one of the largest sources of public R&D investment for hydropower is coordinated by the Water Power Technologies Office of the United States Department of Energy. The three major federal hydropower owners, the US Bureau of Reclamation, the US Army Corps of Engineers, and the Tennessee Valley Authority, also have R&D programs. The U.S. Department of Energy's Hydropower Program budget is typically higher than the EU and it is noteworthy that in the recent past (2016-2021) its annual budget was EUR 99 Million (source: JRC based on IEA^m). According to the Water Power Technologies Office Budget | Department of Energy⁶⁵, the hydropower budget in 2023 was \$59 Million and \$120 Million for marine energy. The average over 2016-2021 was \$33 Million for hydropower and \$77 Million for marine energy. The Bipartisan Infrastructure Law, which passed in 2021, provided an additional \$36 Million for U.S. hydropower R&D to be spent between 2022-2025.

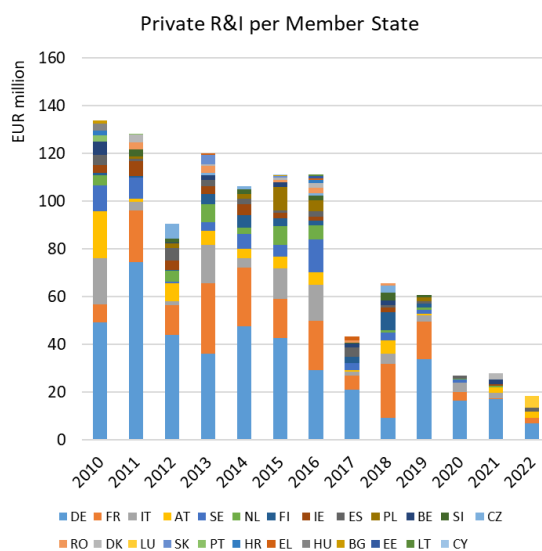
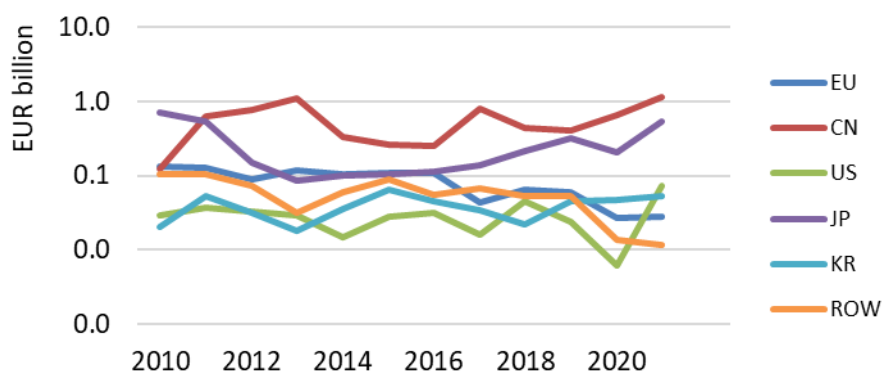
2.6 Private RD&I funding

Private R&I figures are estimated using patenting output as a proxy and should be interpreted with caution; as such they should be viewed as provisional and indicative of a trend rather than reflecting absolute values. Figure 8 shows that China is leading the sector and the global trend follows the Chinese one (data of 2021 are not fully completed). In the EU, Germany, France, Italy, Austria and Sweden are the member states with the largest contribution in 2021. The lead companies from 2015 are VOITH PATENT GMBH (DE), GE RENEWABLE TECHNOLOGIES (FR), MAHLE INTERNATIONAL GMBH (DE), AKTIEBOLAGET SKF (SE), ELECTRICITE DE FRANCE (FR), DIVE Turbinen GmbH Co KG (DE), ANDRITZ HYDRO GMBH (AT), ALSTOM RENEWABLE TECHNOLOGIES (FR), Rehart GmbH (DE), MINESTO AB (SE), EQA PROJECTS BV (NL), HUTCHINSON SA (FR), Forsnetics AB (SE), Neglenco Srl (IT), O TIMONIEMI OY (FI), CR ENERGY SPÅ (PL), ZECO DI ZERBARO E COSTA E C SRL (IT), SOCIETE CIVILE DE BREVETS MATIERE (FR), AQUAKIN GMBH (DE), ENEL INGEGNERIA E RICERCA SPA

^m Data for the US is available up to 2015. After 2015, we have included old IEA calculations/data. The IEA is in the process of revising this data in cooperation with the US authorities.

(IT), FACTORYTALY SRL (IT). Voith Patent GMBH and GENERAL ELECTRIC also appear in the top 20 global list, which is dominated by China and Japan.

Figure 8. R&I investments. Data for 2021 and 2022 are not yet completed.



Source: JRC

Alstom, Andritz Hydro and Voith estimate that together they spend EUR 150 Million annually on R&D in Europe (contract R&D projects included) ⁶⁶.

2.6.1 Venture capital and early and later-stage investments

Two types of investments are here considered.

(1) Early / late stage investments in companies. Those companies are (or have been) start-ups and growth companies. Investments do not reflect expenditures by those companies. See Annex 3 for further details.

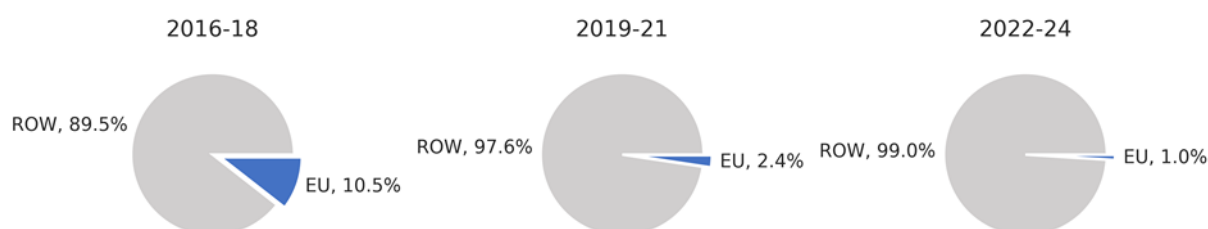
(2) R&D expenditures by the subsidiaries of the largest corporate R&D investors. Expenditures in R&D activities are operational costs and not investments in a company as in (1). Private R&D figures are estimated using patenting output as a proxy.

There is no connection between (1) the early / late stage investments indicators and (2) the private R&D expenditures indicator, and the population of companies addressed in (1) is totally different from the population of companies addressed in (2).

The early and later stages investments in venture capital (VC) companiesⁿ over the considered period are depicted in Figure 9 and Figure 10.

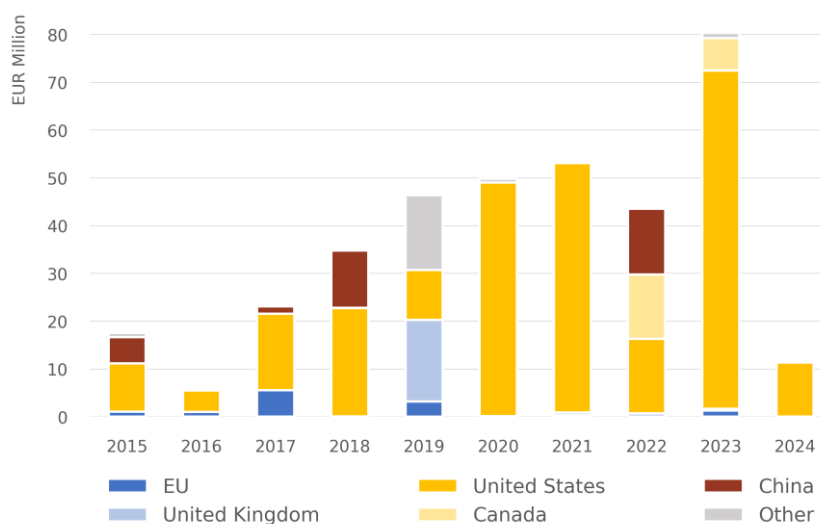
- Global VC investment is driven by US ventures, which have historically captured a large share of the total (74 % since 2019). Lead US companies include Natel Energy, Emrgy and Quidnet Energy.
- With a plummeting number of deals (3 to 4 times less compared to previous years, including in the US), VC investments are EUR 11.3 Million in 2024.

Figure 9. Share of VC/PE investment in the EU and in the ROW for all deals over the 9 years, by period of 3 years. JRC based on Pitchbook.



Source: JRC based on Pitchbook

Figure 10. Global VC/PE investment by region (all deal types)

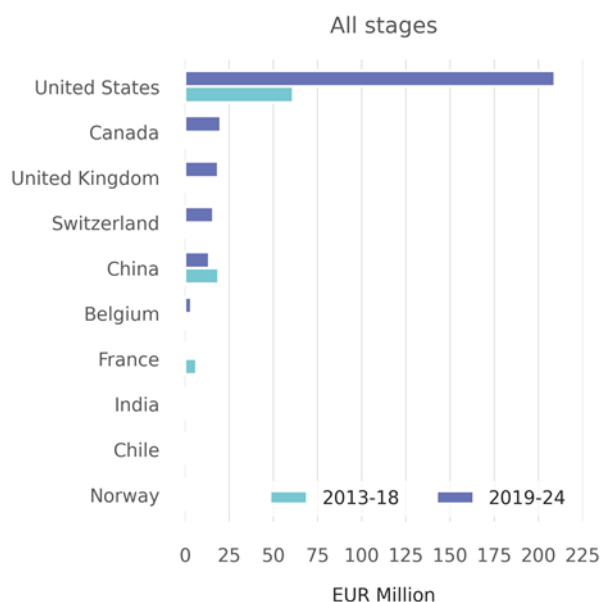


Source: JRC based on Pitchbook

ⁿ Venture capital (VC) is a form of private equity and a type of financing that investors provide to start-up companies and small businesses that have long-term growth potential. Venture Capital companies are companies that have been at some point part of the portfolio of a VC investment firm (or that have received Angel or Seed funding, or are less than 2 years old and have not received funding).

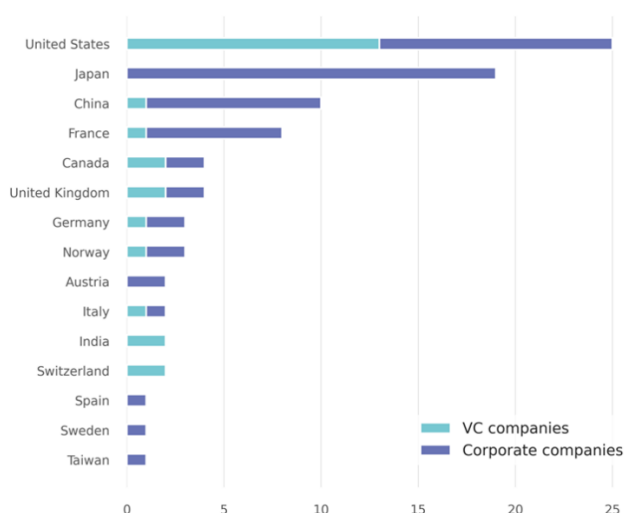
Based on investments, innovative companies can be identified, and Figure 11 shows this data (see Annex 3 for details) aggregated at the country level, while Figure 12 shows the countries with the largest number of innovative companies. The US dominate VC/PE investment, while private investments are lower than the EU's ones.

Figure 11. VC/PE investment in top 10 beneficiary countries, by period for all deals.



Source: JRC based on Pitchbook

Figure 12. Number of active innovating companies by type over a 6-year period, ranking of top 15 countries. [VC companies count over 2019-24. Count of corporate companies over 2017-22].



Source: JRC based on PitchBook and PATSTAT data

2.7 Patenting trends

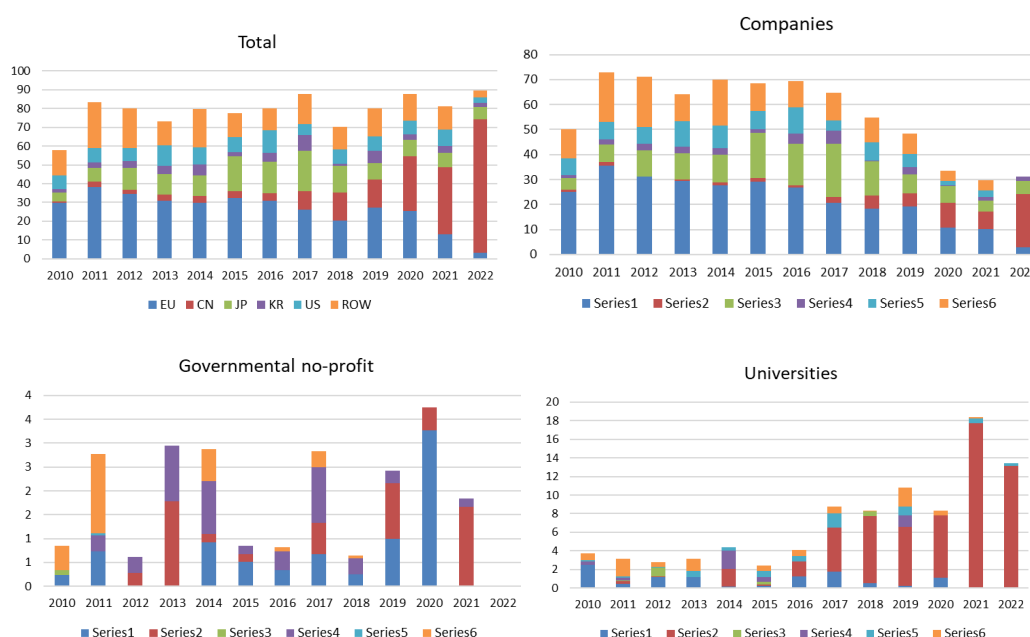
The present patent analysis is based on data available from the European Patent Office EPO PATSTAT 2021b (autumn), using the CPC (Coordinated Patent Classification) code Y02B 10/50 (Hydropower) and Y02E 10/20. Details of the analysis are described in dedicated JRC publications ⁶⁷, ⁶⁸ and in Annex 4. China is the main patent leader (partially also due to the different patenting procedure in the country) while EU, Japan and Korea perform similarly, and better than the U.S. During the period 2010-2022, 696 patents were registered in the EU, 342 were high-value inventions (i.e. applicants of the EU tend to extend patent families to more than one patent office). Within the EU, and considering high-value inventions, Germany is leading, followed by France, Italy and Austria. During the period 2020-2022, 45% of EU inventions were high-value, higher than the 2% of China (but where the total number of inventions was above 6,000). The EU holds 16% of all high-value inventions globally (2020-2022) (Table 5). Therefore, the EU is the lead region, after China, in terms of high-value inventions, although this trend is decreasing over the years, especially due to the Chinese increasing trend (Figure 13 and Figure 14).

Table 5. Number of inventions and share of high-value inventions and international activity (2020-2022). Percentages on the total are also calculated.

	Total	High-value	International	Total	High-value	International
ROW	118	30	28	118	25%	24%
JP	274	23	51	274	8%	19%
KR	310	9	90	310	3%	29%
CN	6277	136	51	6277	2%	1%
US	62	19	34	62	30%	55%
EU	92	41	22	92	45%	24%

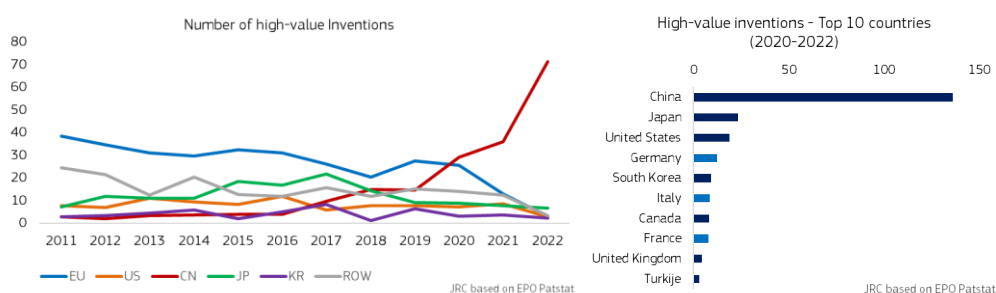
Source: JRC based on EPO Patstat, 2022 data is incomplete.

Figure 13. Patent activity for different owner categories (high-value inventions).



Source: JRC based on EPO Patstat, 2022 data is incomplete

Figure 14. Number of high-value inventions (left) and Top 10 countries (right).

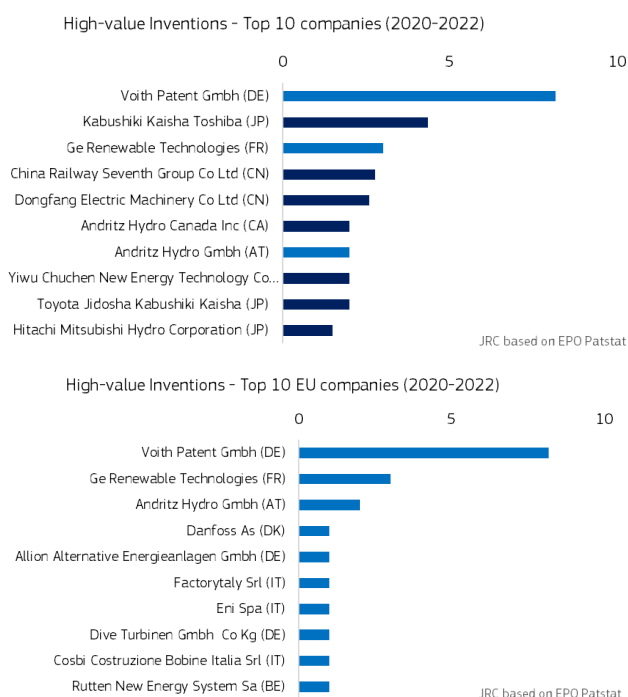


Source: JRC based on EPO Patstat, 2022 data is incomplete

Figure 15 shows the top 10 companies with high-value invention patenting. Three companies from the EU are in this top 10: Voith Hydro, GE Renewable Technologies, and Andritz Hydro, sharing together 49% of high-value inventions from the top 10 leader companies worldwide. It must be noted that patenting activity is not the only indicator to measure innovation, as innovation also includes publications, R&I, exports, participation in research projects and application of new technologies.

R&D expenditures by the subsidiaries of the largest corporate R&D investors, based on patenting activity, are depicted in Figure 8. Expenditures in R&D activities are operational costs and not investments in a company as the late and early-stage investments. The trend is decreasing from 2010, probably due to the increase of R&D activities and related patenting activities in the other renewable energy sectors.

Figure 15. Top patenting companies.



Source: JRC based on Patstat, 2022 data is incomplete

2.8 Scientific publication trend

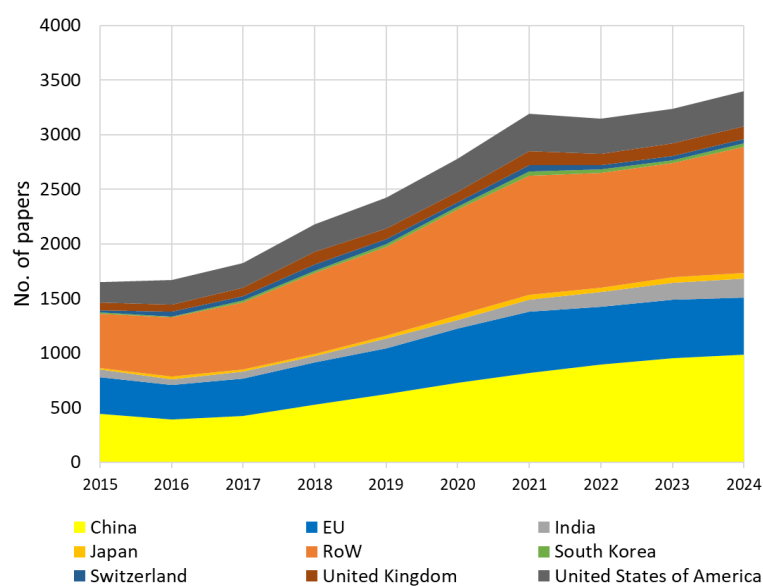
Hydropower research covers a wide range of scientific areas: energy, engineering, environmental and water resource sciences, geology, fisheries and many others. Jiang et al., (2016)⁶⁹ observed a rapid growth rate of publications related to hydropower, highlighting the increasing demand for hydropower-related research (Figure 16). Post construction issues of hydropower development are more attractive for scholars than energy technology itself, and an interdisciplinary trend of hydropower research is emerging from the interaction of natural science, social science and engineering hydropower technology ⁶⁹.

Leading funding agencies for the period 2016-2020 are various National Foundations in China, the National Council for Scientific and Technological Development and the CAPES (Coordenação de Aperfeiçoamento de Pessoal de Nível Superior) in Brazil, the EU (with the H2020 and ERC programs), the NSERC (Natural Sciences and Engineering Research Council) in Canada and the NSF (National Science Foundation) in the US ⁷⁰.

According to Figure 16 and Figure 17, EU's institutions are leading the publication trend, together with China, and this may reflect the leading position of the EU in high-value inventions. In the [2022 CETO report](#), additional keywords were selected to better analyse the bibliometric trends: "dam" ("hydropower" and "dam" have to be found 3 keywords apart or less in the title or in the abstract), "reservoir" ("hydropower" and "reservoir" have to be found 3 keywords apart or less in the title or in the abstract), penstock, turbine, draft tube, fish passage, hydropeaking and sediment (including sedimentation and turbine erosion), as proxy on civil structures, electro-mechanical equipment and environmental operation. The keyword "hydropower", or "hydroelectric" or "hydro electric" must also be present in the text, but with no proximity requirements. Table 7 of the [2022 CETO report](#) shows the bibliometric trend of these keywords. Comparing the EU with the rest of the world, the EU is a lead region in the analysed sub-topics, with an evident leadership in the subtopics hydropeaking, fish passages and sediments. The EU is also well positioned when considering the other keywords, having a publication activity almost similar to the leading country, if the population is taken into account (i.e. number of publications per inhabitant). The main concurrence is exerted by China⁷¹. The percentage of highly cited papers is similar among countries (Table 6).

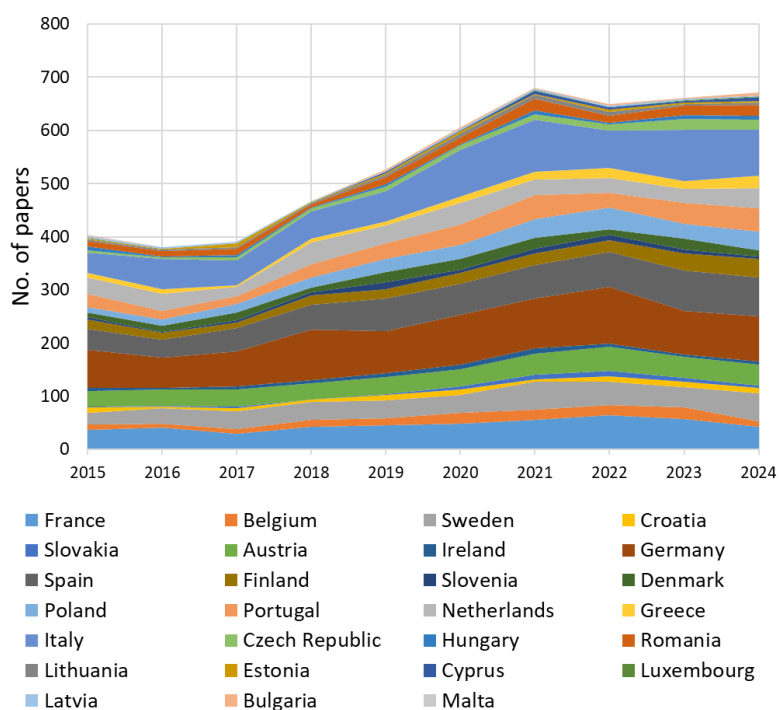
Table 7 lists the first authors, based on number of peer-reviewed publications from 2021 to 2025, according to Scopus ("hydropower" or "hydroelectric" in the keywords or title or abstract), until reaching EU ones. Only two EU authors (including the first author of this report), are in the list. Hydropower research covers a wide range of scientific areas, thus the results here presented, based on a few hydropower-related keywords, do not represent the full database, and are only indicators.

Figure 16. Number of scientific papers published in peer-reviewed journals.



Source: JRC

Figure 17. Number of scientific papers published in peer-reviewed journals for EU member states.



Source: JRC

Table 6. Highly cited papers on hydropower (top 10% cited normalised per year and field, 2015-2024).

	Documents total	Highly cited documents	Highly cited documents (%)
RoW	8525	856	10.0%
China	6784	805	11.9%
EU	4447	555	12.5%
United States of America	2781	380	13.7%
United Kingdom	993	117	11.8%
India	985	158	16.0%
Switzerland	435	60	13.8%
Japan	322	34	10.6%
South Korea	227	23	10.1%

Source: JRC

Table 7. List of main authors and number of papers 2021-2025 (access: June 2025), under the keyword Hydropower (source: Scopus): global analysis (left), European (EU+UK+NO+CH) analysis (right). The analysis does not consider weights like the position of the authors in each publication, e.g. first author contribution.

Author	N° of papers	Author	N° of papers
Cheng, C.	107	Quaranta, E.	28
Liu, P.	62	Ramos, H.M.; Boes, R.M.	27
Chen, D.	41	Vagnoni, E.; Chen, D.	21
Liao, S.	39	Kuriqi, A.; Hayes, D.S.; Dahlhaug, O.G.; Albayrak, I.	20
Ming, B.	38	Geist, J.; Castelletti, A.	18
Liu, B.	37	Jurasz, J.	17
Zhou, J.; Wang, Z.; Shen, J.	35	McNabola, A.; Hauer, C.	16
Zheng, Y.; Wang, Y.	34	Xu, B.; Schmutz, S.; Pérez-Sánchez, M.	15
Zhang, J.; Chang, J.	31	Panthi, K.K.; Pander, J.; Nicolet, C.	15
Zhao, Z.; Yang, W.	29	Münch-Alligné, C.; Boavida, I.; Aufleger, M.	15
Quaranta, E.; Zheng, Y.	28		
Ramos, H.M.; Boes, R.M.	27		

Source: Scopus

2.9 Assessment of R&I project developments

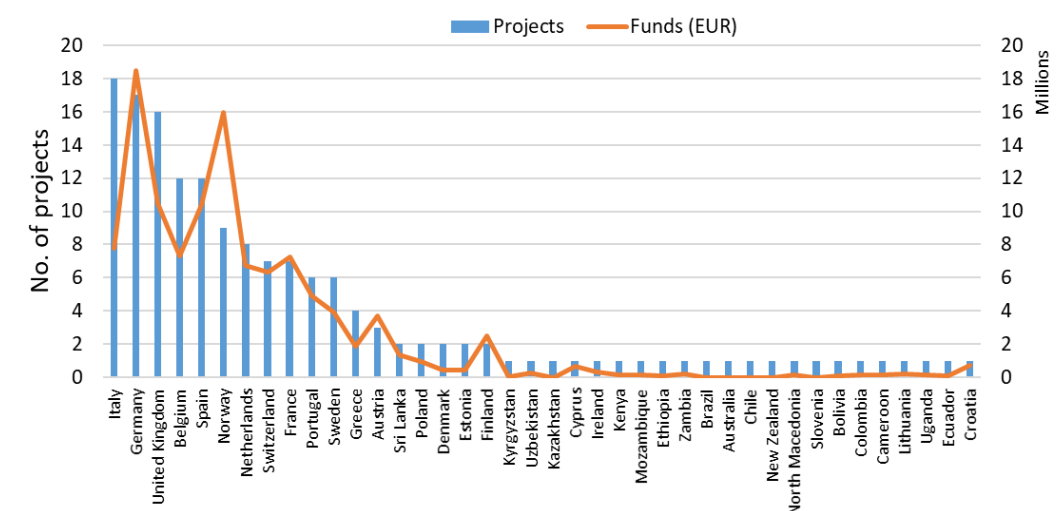
Horizon Europe (Horizon Europe, 2025) is the EU's key funding programme for research and innovation under the EU Research & Innovation Framework Programmes. Following the Multiannual Financial Framework Midterm Review (MTR) decision, the indicative funding amount for Horizon Europe for the period 2021-2027 is EUR 93.5 Billion. It tackles climate change, helps to achieve the UN's Sustainable Development Goals and boosts the EU's competitiveness and growth. The programme facilitates collaboration and strengthens the impact of research and innovation in developing, supporting and implementing EU policies while tackling global challenges. The European Union has funded several projects in the field of hydropower under the Horizon Europe programme. From 2015 to 2024, 42 projects have been funded, with a peak in 2018 (10 projects). Figure 18 depicts the received EU contribution for each country, and the total number of projects between

2015 and 2022. UK, Belgium and the Netherlands, that are flat countries, have received many funds, and this is associated mainly to the development of low head hydropower equipment. Italy and Germany are amongst the most funded countries, since their territory is suitable both for low head hydropower and for high head plants in the Alpine environment. Some non-EU countries have been involved, and, in particular: Uzbekistan, Kyrgyzstan, Kenya, Zambia, Mozambique, Ethiopia, Chile, Australia, Brazil, New Zealand, North Macedonia, Colombia, Uganda, Cameroon, Ecuador, Bolivia. Norway and Switzerland are also very active partners.

Table 8 lists some of the Horizon funded projects. The three most recurrent topics are hidden and low head hydropower, environmental mitigation and increase of flexibility. It is worth mentioning the research projects Hydropower Europe (now, ETIP Hydropower) and PEN@Hydropower, bringing together stakeholders to improve networking in the EU hydropower sector, supporting sustainable hydropower.

Recent Horizon calls deal with hidden hydropower opportunities^o (see also Quaranta et al., 2022, typically associated to low head hydropower), hydropower digitalisation^p (associated to flexibility, Quaranta et al., 2021, Kougias et al., 2019, Quaranta et al., 2023⁷⁵), pumped-storage hydropower^q and sustainable refurbishment^{r, s}. The EU is also active member of the Hydropower TPC of IEA^t. The Joint Programme Hydropower^u of the European Energy Research Alliance (EERA) supports the role for hydropower as an enabler for the renewable energy sector. The exploratory activity SustHydro, coordinated by the Emanuele Quaranta at the JRC, aimed at identifying sustainable and hidden hydropower opportunities in the EU to support the previous activities and EU's policies and trying to support policy development with evidence-based facts, connecting industrial knowledge with science (see e.g. Quaranta et al., 2021, Quaranta et al., 2022).

Figure 18. Received funds and number of projects by country (Interreg projects not included), 2015-2022.



Source: JRC

^o <https://ec.europa.eu/info/funding-tenders/opportunities/portal/screen/opportunities/topic-details/horizon-cl5-2021-d3-03-11>

^p <https://ec.europa.eu/info/funding-tenders/opportunities/portal/screen/opportunities/topic-details/horizon-cl5-2022-d3-03-08>

^q <https://ec.europa.eu/info/funding-tenders/opportunities/portal/screen/opportunities/topic-details/horizon-cl5-2024-d3-01-16>

^r <https://ec.europa.eu/info/funding-tenders/opportunities/portal/screen/opportunities/topic-details/horizon-cl5-2023-d3-02-09>

^s <https://ec.europa.eu/info/funding-tenders/opportunities/portal/screen/opportunities/topic-details/horizon-cl5-2024-d3-01-07>

^t <https://www.ieahydro.org/work-programme/>

^u <https://www.eera-hydropower.eu/>

Table 8. List of Horizon projects (2015-2024) and number of partners.

Starting date	Title	Total cost of the project (EUR).	Project acronym	Partners
2024	Revolutionary refurbishment for an efficient and eco-friendly hydropower	3999951	RevHydro	8
2024	new Solutions for Hydropower plants to Enhance operational Range, Performance and improve environmental impAct	3861651	SHERPA	7
2024	Innovative storage technology and operations in hydropower	7996175	STOR-HY	15
2024	Demonstration of sustainable hydropower refurbishment	7 506 337	REHydro	16
2024	Novel long-term electricity storage technologies for flexible hydropower	4315796	Store2Hydro	8
2023	Digital solutions for improving the sustainability performance and FLEXibility potential of HYDROpower assets	4 038 519	D-HYDROFLEX	7
2023	DIGITAL MAINTENANCE FOR SUSTAINABLE AND FLEXIBLE OPERATION OF HYDROPOWER PLANT	4000000	Di-Hydro	7
2023	intelligent Asset Management Platform for Hydropower operation and maintenance	4100000	iAMP-Hydro	7
2022	Hidden Hydro Oscillating Power for Europe	4854230	H-HOPE	9
2021	Hydropower For You - Sustainable small-scale hydropower in Central Asia	11488428	Hydro4U	12
2020	Augmenting grid stability through Low-head Pumped Hydro Energy Utilization & Storage	4996825	ALPHEUS	6
2020	An innovative axial turbine for conversion of hydro-kinetics energy to electricity in rivers and canals	2192125	HyKinetics	2
2019	Predicting changes in species interactions following species loss in hydroelectric reservoir islands	183454	Lost-Biodiv	1
2019	HYdro-POwer-Suite	2397120	HYPOS	5
2020	Rlver flow regulation, fish BEhaviour and Status	4048220	RIBES	4
2019	Interoperable solutions for implementing holistic FLEXibility services in the distribution GRID	8544133	FLEXIGRID	6
2019	A revolutionary HYDRO POWER technology to sustainably exploit super-low-head water steps	3511500	Turbulent	1
2019	Hydropower solutions for developing and emerging countries	2938374	HYPOSO	13
2019	Hydropower Extending Power System Flexibility	18162950	XFLEX HYDRO	7
2019	Waterjade: the global platform to predict water resources	71429	Waterjade	2
2019	A scalable and sustainable grid-scale energy storage system utilising 3rd generation flywheel technology for effective integration of renewable energy.	3465625	Teraloop EES	2
2019	Active Flow Control system FOR improving HYDRaulic turbine performances at off-design Operation	4711589	AFC4Hydro	6
2018	Energy from water in motion: efficient, customisable off-grid hydro-electricity for rural areas with stream access	71429	HYDROGO	2
2018	HYDROPOWER-EUROPE	993571	HYDROPOWER-EUROPE	4
2018	Kinetic Micro Hydro System for electrification of rural areas.	71429	SMART Slowflow	2
2018	A highly scalable grid-scale energy storage system utilising 3rd generation flywheel technology for effective integration of renewable energy.	71429	Teraloop ESS	2
2018	Statistically combine climate models with remote sensing to provide high-resolution snow projections for the near and distant future.	180277	ClirSnow	2
2018	Increasing the value of Hydropower through increased Flexibilty	5716989	HydroFlex	5
2018	An innovative axial turbine for conversion of hydro-kinetics energy to electricity for river applications.	71429	HyKinetics	2
2017	A EUROpean training and research network for environmental FLOW management in river basins	3923989	EUROFLOW	8
2017	Environmentally efficient full profile drilling solution	2811875	ECO-DRILLING	2
2017	Water flow kinetics energy exploitation for mini/micro hydropower plants.	71429	RIVER-POWER	2
2017	Development of a novel wave tidal energy converter (WATEC) to lower renewable electricity generation costs.	71429	WATEC	1
2017	Quantifying ecological effects of small hydropower in Alpine stream ecosystems	180277	SHYDRO-ALP	2

2016	A GLOBAL MOVEMENT FOR ENVIRONMENTAL JUSTICE: The EJAtlas	1910811	EnvJustice	2
2016	Fishfriendly Innovative Technologies for Hydropower	7171550	FITHydro	9
2016	DAFNE: Use of a Decision-Analytic Framework to explore the water-energy-food NExus in complex and trans-boundary water resources systems of fast growing developing countries.	5420223	DAFNE	11
2016	Adaptive Management of Barriers in European Rivers	6238104	AMBER	10
2016	Knowledge Exchange for Efficient Passage of Fishes in the Southern Hemisphere	135000	KEEPPISH	8
2015	PROFITABLE LOW HEAD HYDROPOWER	1512893	Hydrolowhead	2
2015	An online software platform for effectively matching investors with start-ups, focused on boosting the cleantech sector	71429	CAGIX	2
2015	Bringing INnovation to onGOing water management – A better future under climate change	7822423	BINGO	6

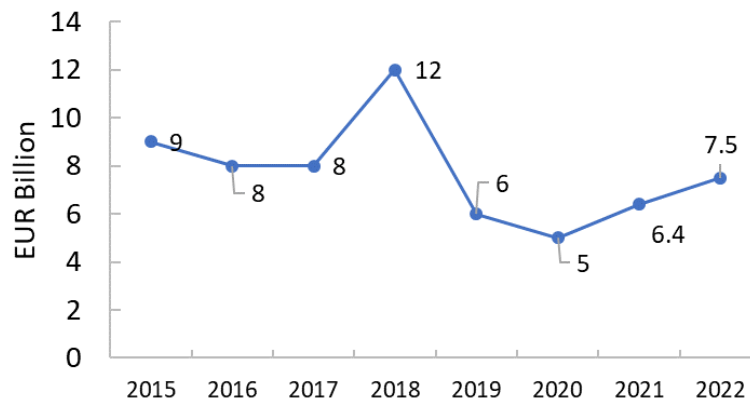
Source: JRC

3 Value chain Analysis

3.1 Turnover and Gross Value Added

The annual turnover of hydropower electricity generation in the EU is depicted in Figure 19⁷². In 2022, leading member states were France (EUR 0.56 Billion), Italy (EUR 0.95 Billion), Austria (EUR 1.1 Billion) and Germany (EUR 1.1 Billion). According to 2021 data, European hydropower manufacturers spend more than 5% of annual turnover on R&D, which is more than twice the European industry average⁷⁰. The greatest turnover was approximately EUR 12 Billion in 2018. Turnover is increasing from 2020. However, according to another sources (Prognos, 2024), the real dimension of turnover could be much higher, going beyond EUR 100 Billion, if including also civil works in addition to the equipment.

Figure 19. Hydropower turnover in the EU, from EurObserv'ER (2024 data).

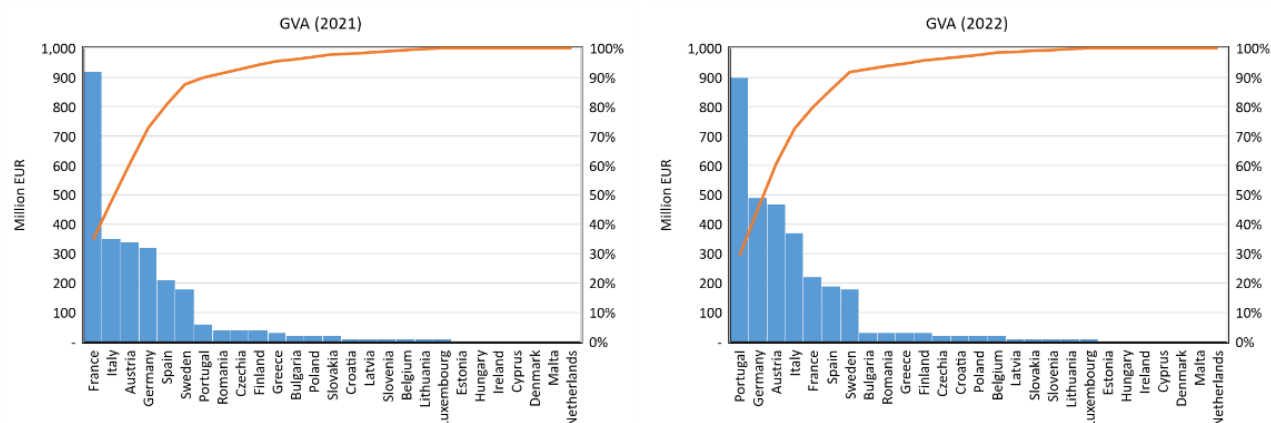


Source: EurObserv'ER

EurObserv'ER provides an estimate of Gross Value Added (GVA) based on a uniform modelling methodology for all technologies (Figure 20). The estimate is based on the recorded money flows into 1) new installations; 2) O&M of existing capacity; 3) production (section 3, EUR approx. 800 Million in 2023) and trade of equipment (approx. EUR 250 Million in 2024 for extra-EU trade); 4) production and trade of biomass (not relevant for hydro), which are converted to GVA using different technology specific factors – both direct and indirect effect is included, but not induced effect. Gross value added is the total value of output in the sector minus the value of intermediate consumptions and therefore, used as a measure of the contribution to GDP.

The GVA of the hydropower sector, according to EurObserv'ER, in 2021 and 2022 is depicted in Figure 20: the EU's GVA was EUR 2.72 Billion in 2021 (EUR 1.95 Billion in 2020) and EUR 3.14 Billion in 2022, with an increasing trend. According to Prognos analysis, the GVA in manufacturing in 2021, 2022, 2023 was EUR 210, 180, 250 Million, in construction and operation it was EUR 22.6, 18.4, 18.7 Billion, and in Research and Development it was EUR 29, 13 and 14 Million, respectively. New EurObserv'ER's data in 2025 shows an increase of EUR 930 Million in 2023 with respect to 2022, but the new 2022 data is EUR 1,310 Million instead of EUR 3,140 Million. According to the author's judgement, Prognos results are more stable and complete, as they tend to capture a larger portion of the hydropower sector.

Figure 20. Gross Value Added (GVA) for each EU member state in 2021 (left) and 2022 (right), EurObserv'ER's data.



Source: EurObserv'ER

Hydropower contribution to the economy also includes profits on energy generation and ancillary services, that range between EUR 20 and 30 Billion per year. Hydropower contribution to EU+UK GDP is expected to increase considerably by 2030 and exceed EUR 40 Billion or even reach EUR 50 Billion, depending on the renewable scenario^v evolution (Diversified Supply or Reference Scenario, respectively)⁷³.

According to a 2015 study, as no more recent data have been found, in the whole of Europe, the annual value creation^w was approximately EUR 38 Billion in 2015, which may grow to some EUR 75 Billion to 90 Billion by 2030. Direct tax contributions are estimated at almost EUR 15 Billion annually, or more than one third of total value creation, which is several times more than the limited volume of subsidised payments to small hydropower. A substantial share of this value goes directly to local and regional budgets and helps to foster regional development. However, as energy generation from hydropower is almost stable, the actual situation might not be so different.

Whilst it is difficult to estimate the associated benefits, the multi-purpose functions of hydropower represent an additional annual economic value of EUR 10 Billion to 20 Billion, even when neglecting the potential value of avoided damages from flood events, which may be substantial. These benefits can be expected to increase in the future, for instance due to an increased need for water management and flood control. IHA estimates that through the water storage function of its reservoirs, the hydropower industry prevents US\$131.3 Billion in annual GDP losses from drought incidents (IHA, 2024). During the past 30 years a flood year increased, on average, the annual EU hydropower capacity factor by +7% more compared to non-flood years. In years with storms the annual EU hydropower increased by +5.8%, while droughts/heatwaves lowered it by -6.5%⁷⁴.

^v The scenario 'Diversified Supply Technologies' follows the EU's long-term decarbonisation pathway and uses a mix of different technologies, including RES. It achieves a significant reduction of carbon emissions in the power sector (> 95% by 2050) and assumes a strong growth of renewables, mainly wind power. In contrast, the "Reference" scenario reflects a more conservative development scenario that fails to meet the carbon reduction targets by 2050.

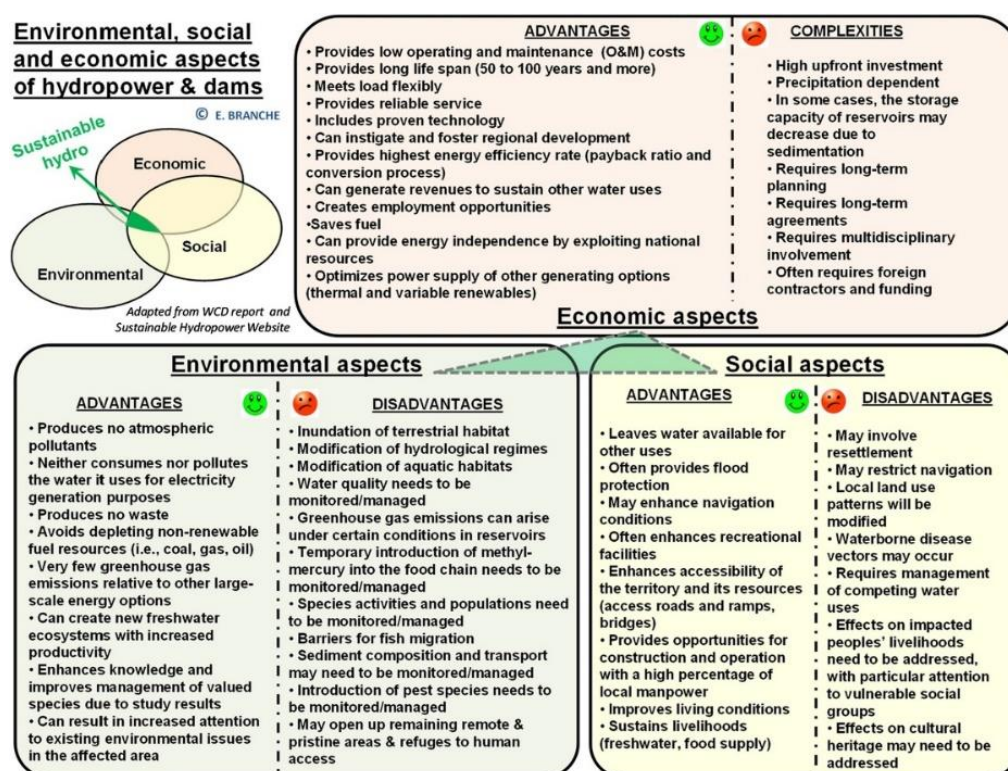
^w Value creation comprises of the contribution of different economic sectors to gross domestic product (GDP).

3.2 Environmental, social and economic sustainability

Large hydropower plants, and large reservoirs in general, are complex systems, and they interact with the hydrosphere, the biosphere, the lithosphere, the anthroposphere and the atmosphere, with impacts and benefits⁷⁵. Therefore, sustainable development is a major challenge for hydropower, because sustainable hydropower requires attention to a wide range of economic, social and environmental objectives. Hydropower development, especially the large projects, involves several stakeholders.

Overall benefits and impacts are summarized in Figure 21, while Annex 1 summarizes the sustainability indicators of hydropower selected for the CETO project. More information in the 2024 report.

Figure 21. Benefits and impacts of hydropower.



Source: Branche, 2017⁷⁶

In order to consider the transversal impacts and benefits of hydropower, the Hydropower Sustainability Alliance (HSA) created the Hydropower Sustainability Standard — a multi stakeholder certification system that evaluates projects' environmental, social and governance performance. The Standard builds on the earlier Hydropower Sustainability Assessment Protocol, originally developed by the International Hydropower Association, and sets global sustainability expectations for hydropower. As of September 2025, dozens of projects worldwide have been assessed, and Europe has seen its first Gold certifications. The ones listed in Table 9 are in Europe. There are also some projects aimed at increasing biodiversity (e.g. by means of afforestation/regeneration/maintenance programs) at hydropower plants in Europe⁷⁷.

Table 9. Hydropower projects in Europe which have been assessed against the Hydropower Sustainability Tool.

Country	Project name	Installed capacity	Assessment tool & certification
Iceland	Fljótsdalsstöð (Kárahnjúkar)	690 MW	HSS – Gold Certification
Iceland	Hvalárvirkjun	55 MW	HSS – Public Consultation
UK	Coire Glas	1 500 MW	HSS – Gold Certification
France*	Romanche-Gavet	94 MW	HSAP
Germany*	Walchensee	124 MW	HSAP
Sweden	Semla IV	3.5 MW	HSAP
Albania	Devoll	256 MW	HSAP
Austria*	Kaunertal	1 015 MW	HSAP
Croatia*	Program Sava	151 – 156 MW	HSAP
Iceland	Blanda	150 MW	HSAP
Iceland	Hvammur	82 MW	HSAP
Norway	Jostedal	288 MW	HSAP

** in the EU. Source: IHA.*

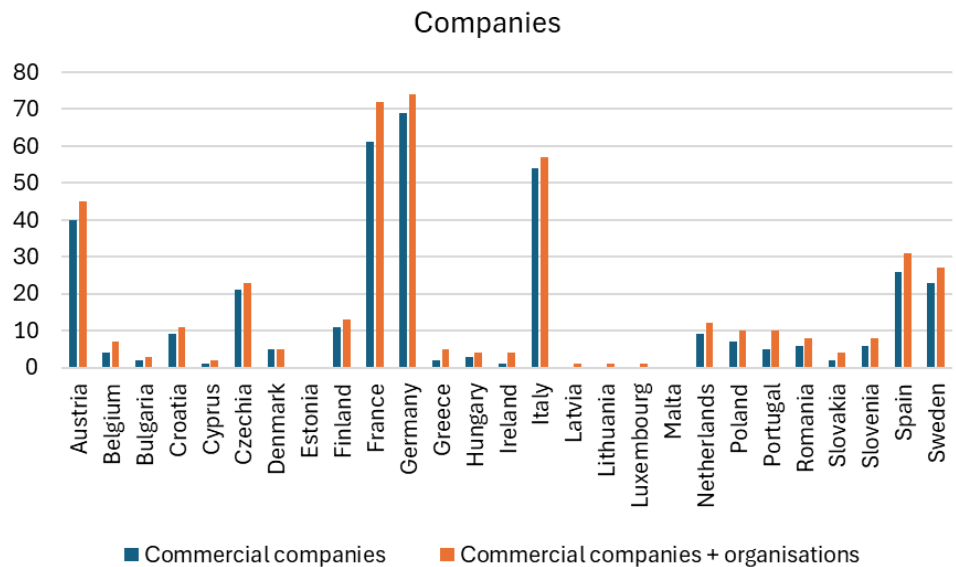
3.3 Role of EU's Companies

According to Prognos, slightly more than 17,100 enterprises in the EU are active in the hydropower sector: 48 for R&D, 45 in manufacturing, 63 in construction and 17,000 in operation. According to EREF (European Renewable Energy Federation), the EU's small hydropower sector consists of more than 4,500 innovative enterprises (mainly SMEs) employing more than 60,000 professionals and generates an annual turnover of around EUR 3 billion. However, tracking and listing them is not easy, and the previous data are mainly based on estimates.

According to the author knowledge, the largest sample of hydropower companies is available in the Hydropower&Dams database of Aqua Media International. Based on this database, Figure 22 depicts the number of hydropower and dam organisations (associations, policy institutes) and companies (e.g., manufacturers, operators, engineering companies) in EU Member States (website accessed in August 2025) ⁷⁹. This database is not exhaustive as it reflects the ones which registered to appear on the website. The database highlights that the main hubs of hydropower activity are in France, Germany and Italy, and that certain countries such as Austria, Spain, Sweden, and Czech Republic host a considerable number of them. Out of total 438 entries, 84% are commercial companies, the others are national and international organisations. Figure 23 shows a non-exhaustive map of European R&D centres of hydro equipment manufacturers.

Manufacturing capacity of EU hydro companies is quite large and a flagship for the EU. EU companies contribute by approx. 40% to the global extra-EU export of hydropower equipment. Many European engineering and consultancy companies offer knowledge, expertise, or consulting to hydropower projects in and outside of Europe. Hydropower does not use critical material, so that manufacturing does not depend on third countries (e.g. China). EU companies also hold several patents and high-value inventions, already discussed in previous chapters. Since they first appeared, turbines manufactured in Europe have spread all over the world, and it is not wrong to state that it was Europe that gave light to the world (ESHA, 2001)⁷⁸.

Figure 22. Leading hydropower and dam engineering organisations/companies per EU's Member State ⁷⁹.



Source: Hydropower&Dams website, 2025, supplier guide.

Figure 23. Non-exhaustive map of European R&D centres of hydro equipment manufacturers ⁸⁰.



3.4 Employment

The deployment of renewable energy generates several employment opportunities in different sectors, with different levels of qualification and duration. It is possible to distinguish between direct, indirect and induced jobs. Direct employment refers to employment generated directly by core activities (implementation value chain). Indirect employment includes jobs in upstream industries that supply and support core activities (e.g., for sustainability, the climate adaptation value chain). Induced employment (generated by productive end uses) encompasses jobs resulting from additional income being spent on goods and services in the broader economy, such as food, clothing, transportation and entertainment (utilisation value chain). Employment can be measured in different ways. Full-time equivalent jobs are equal to one person working full-time over the course of a year. Person-days reflect the amount of work done by one person working full-time for one day.

Employment in hydropower industry spans various value chain elements as project design, manufacturing, project construction and O&M. The sector employment generally includes engineers, geologists, ecologists, economists, technicians, and skilled workers. It also provides employment to scientists, as well as a wide range of scientists working in corporate and academic R&D activities. On the basis of existing installations, IRENA estimates that every 1 MW of community-owned hydropower installed generates ten full-time equivalent jobs in every year of its operation. This is significantly more than other generation technologies, including the next-best performer, community-owned solar photovoltaics at around three full-time jobs per 1 MW installed. Together, the feasibility, planning and procurement, manufacturing, installation and connection, operation and maintenance, and decommissioning of a small-scale hydropower plant require more than 17,000 person-days for a pico hydro plant (average 5 kW), around 64,000 person-days for a micro hydro facility (50 kW) and over 160,000 person-days for a mini hydro system (500 kW). The labour requirements vary across the value chain. Operation and maintenance work is needed throughout the lifetime of a system – estimated at around 40 years – and therefore represents a large chunk of the labour required (94%, 87% and 78% of total person-days, respectively, for pico, micro and mini hydro facilities). Installation and connection and manufacturing are the next-largest shares (up to 4%, 11% and 15%, for pico, micro and mini hydro facilities, respectively). Decommissioning is included, but its share is negligible⁸¹.

Globally, IRENA calculated that approximately 2.36 Million people worked directly in the hydropower sector in 2021, the largest number in the last monitored decade. Only bioenergy (3.44 Million) and photovoltaics (4.29 Million) exhibited a higher employment level across clean electricity generation technologies. Globally, almost two-thirds of these jobs were in manufacturing, 30% in construction and installation activities and about 6% were in O&M services. China was the largest contributor to hydropower direct jobs, accounting for 37% of global employment. India accounted for about 18% of global hydropower employment, followed by Brazil, Viet Nam, Pakistan, the United States, the Russian Federation and Colombia. In 2021, Ethiopia climbed to ninth place amongst hydropower employers, reflecting the construction of large new structures, such as the Grand Ethiopian Renaissance Dam, the largest hydropower project in Africa. Canada rounded out the top ten ⁸².

The numbers associated to the employment vary depending on the source. According to a study conducted by Prognos AG, Berlin and COWI SE, Kopenhagen, commissioned by the European Climate, Infrastructure and Environment Executive Agency (CINEA), in 2023 in the EU there were 102,000 employed people in the manufacturing, construction and R&D. Alpine countries – i.e. Austria, France, Italy and Germany – were the most relevant employers. According to EurObserv'ER, the number of direct jobs of hydropower ranges between 74,000 and 87,000 ⁷² in 2019, while 89,000 was estimated in 2021 by IRENA⁸², 7.2% of the direct employment in the renewable energy sector, with almost another 30,000 jobs created in external services of hydropower ^{70, x}. A 10% increase of

^x Employment and turnover figures based on EurObserv'ER include also construction and consulting services, which in the case of hydro represent a bigger share of CAPEX than manufacturing indeed. Data are based on modelling based on

hydropower in the year 2030 would create 27,000 jobs in the EU, mainly outside the hydropower sector itself⁷³. According to some recent EurObserv'ER's estimates, direct and indirect jobs were 49,000 and 79,000 in 2021 and 2022, respectively. The 2025 data suggests an increase from 2022 to 2023 of 16,000 professionals, but repropose a value of 25,600 instead of 79,000 in 2022. According to the European Renewable Energies Federation (EREF), 4,500 SMEs are present in the EU on small hydropower, employing approx. 60,000 people⁸³.

Civil, hydraulic, mechanical and electric engineers, hydrologists, are the most typical profiles. Basic knowledge of hydraulic machines, hydraulic structures as well as a good understanding of electric design and grid stability are also important aspects of an educational background. Hydropower projects must adhere to environmental regulations and mitigate potential ecological impacts. Environmental specialists, often environmental engineers, play a crucial role in conducting impact assessments, designing fish passage systems, biotope restoration and ensuring compliance with environmental standards, especially in view of the concession renewal. Technical skills including welding are also vital: having sufficient simple grinders and welders could become an issue in the future as there are fewer people going into manual jobs and automation processes are difficult and expensive. Other skills necessary for the hydropower sector are those related to project management – contracts, asset management, energy, supervision for installation as well as operations and maintenance (O&M). Skills that will become more important over time include data and digitalisation specialists who can incorporate digitalisation practices into hydropower system design. Electric engineers with a high voltage background are also decreasing over time, so this skill set must be fostered.

Additionally, it will be even more important in the future to encourage multidisciplinary approaches, as hydropower is incorporated into the grid and new developments will have multi-purpose use. The collection and analysis of data from sensors and monitoring systems are becoming integral to optimise hydropower plant performance. Data analysts skilled in statistical analysis, machine learning, and programming help extracting insights and improving operational efficiency. Professionals specialising in energy policy and regulations will be instrumental in shaping policies, incentives, and frameworks for hydropower development. They will analyse market trends, assess policy implications, and contribute to the growth of the industry⁸⁴. Empowering women and girls and closing gender gaps is also critical to realizing full equity in many sectors of energy industry, including hydropower. The sector could offer a range of direct and indirect job opportunities for women. However, shares are still lingering around 25% of women, depending on the hierarchical level. With institutional demands targeting improved gender balance, the European Cooperation in Science and Technology (COST) Actions has started to introduce the Gender and Equity Plan (GEP) with an extra officer for this plan in each Action in 2025. The EU COST 21104 Pan-European Network for Sustainable Hydropower (PEN@Hydropower) has launched the Action's Women's Mentoring. At the global scale the "Women in Hydropower Mentoring Programme" is running to tackle challenges that women face in the hydropower sector.

3.5 Energy intensity and Labour productivity

According to data discussed in Figure 37, hydropower schemes present the largest Energy Return of Investment. Despite the very high installations costs of the large ones, mainly due to civil works, capacity factors of hydropower are higher than on-shore wind and photovoltaics, and can also provide flexibility and additional benefits in the water management sector. Overall efficiency can also be above 90%. For more information on employment refer to section 3.4.

CAPEX investments with resulting jobs and turnover effects being assigned to the year of commissioning the project. This is why, in the case of hydro, there are big variations between the years.

3.6 EU Production Data

Hydraulic turbines are important components and a reliable proxy of the investment as they define the power capacity of the station. The market of large-scale units (above 10 MW) is dominated by a small number of companies, while many turbine manufacturers exist in the EU and globally, the majority of which focuses exclusively on small-scale turbines. This section focuses exclusively on the global market of large turbines that are typically hosted in projects worth several EUR hundred Million (or even EUR multi-Billion investments). In monetary terms, such investments represent a large share of the global hydropower market. Besides, the small-scale market is not systematically monitored. An additional particularity of the hydropower market is that a significant part of investments is not monitored as it refers to the civil works and the associated consultancy services. To monitor the trends in hydropower technology development, Prodcom codes for the production of hydraulic turbines and their parts have been selected on Eurostat database (Table 10). For monitoring production in hydropower, the selection of Prodcom codes is based on the Net-Zero Industry Act (NZIA) list of components^y. This includes turbines (28112200 - Hydraulic turbines and water wheels) and their parts (28113200 - Parts for hydraulic turbines and water wheels (including regulators)). However, these codes do not distinguish between turbine size or application.

Table 10. Selection of Prodcom codes as a proxy for production of hydropower technologies.

Prodcom code	Description	Alias
28112200	Hydraulic turbines and water wheels	turbines
28113200	Parts for hydraulic turbines and water wheels (including regulators)	parts

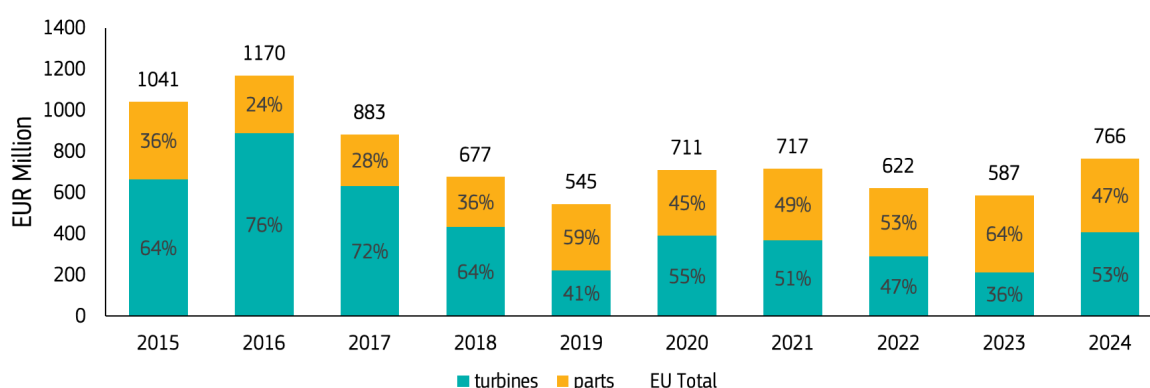
Source: Prodcom

In 2024, the total EU production value of hydraulic turbines and parts rose by nearly one-third compared to 2023, reaching approximately EUR 766 Million. This growth was driven by a 90% increase in turbine production value, offset by a 4% decline in parts production. Germany and Austria were the leading EU producers, accounting for 29% and 27% of total 2024 production, respectively, followed by Italy (19%). Italy’s 2024 production focused on turbines (84%), Austria’s on parts (95%), and Germany’s on turbines, though parts production data for Germany were not disclosed. Over the past decade (2015-2024), the overall EU production value of these commodities decreased by 26%, with a 39% decline for turbines and a 5% decline for parts. Historically, turbines constituted 56% of EU production on average, while parts accounted for 44%. Notably, this trend reversed in the past three years (2022-2024), with parts production rising to represent 55% of the EU total on average (Figure 24).

According to Prognos, production in the category “manufacturing” was EUR 710, 650 and 840 Million in 2021, 2022 and 2023, respectively, which are In line with the value of Production abovementioned, while EUR 93.4, 73.3 and 75.6 Billion in the “Construction and Operation” and EUR 60, 28 and 29 Million in “Research and Development”. Prognos results consider more aspects and should be regarded as more complete.

^y C(2025) 9034 final, https://single-market-economy.ec.europa.eu/document/download/860c257b-16db-429e-91c1-a30bea57e7ce_en?filename=C_2025_9034_1_EN_ACT_part1_v8.pdf

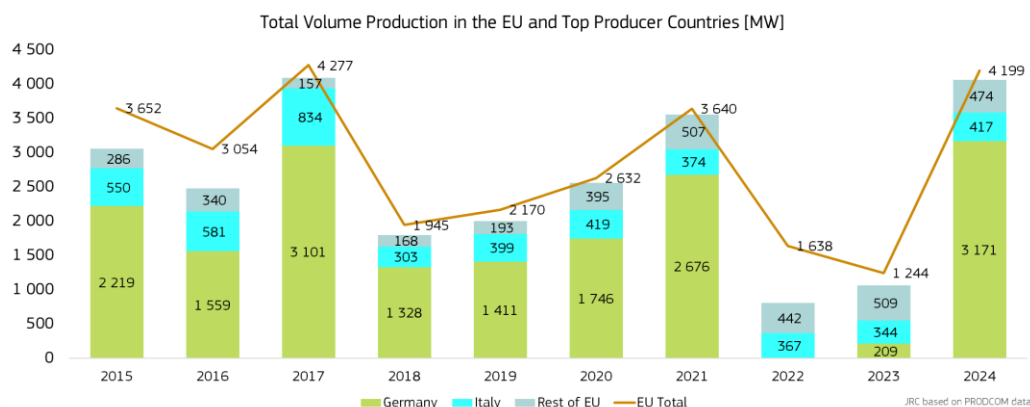
Figure 24. EU production value per commodity [EUR Million].



Source: JRC based on PRODCOM data.

In terms of power (MW), the Member States had disclosed data only for the production of turbines. Figure 25 illustrates the EU hydraulic turbines production (only 28112200) in quantities (MW). The sum of countries' production (boxes) is lower than the 'EU Total' (line) because some Member States keep their production data confidential. However, Eurostat includes confidential data in the 'EU Total' estimates. In 2024 the hydraulic turbines production more than tripled compared to 2023, reaching approximately 4.2 GW. Germany and Italy were the leading EU producers, holding 76% and 10% of the EU 2024 production, respectively, followed by Finland (9%). Over the past decade (2015-2024), EU production grew by 15%, with an annual compound growth rate of 1% and an average annual value of around 2.8 GW.

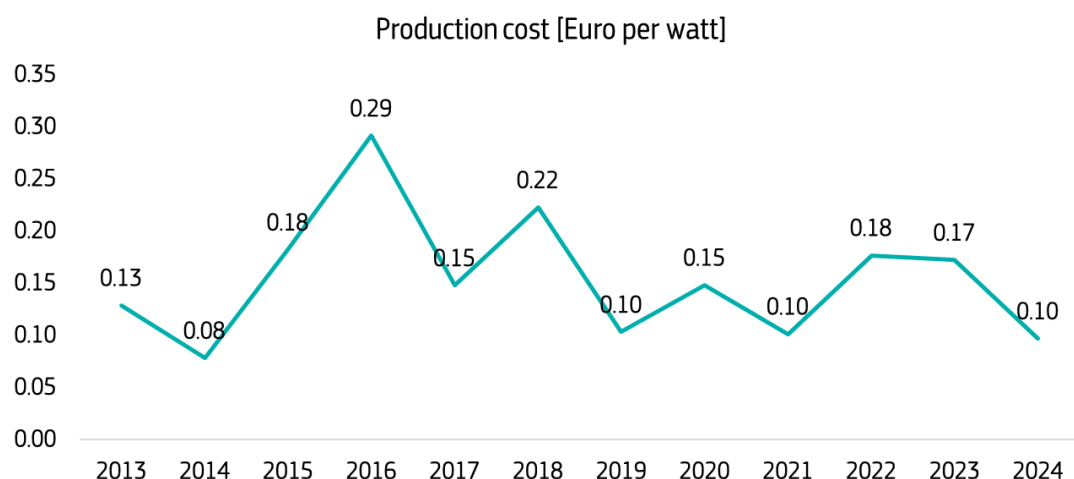
Figure 25. EU production value per commodity [MW]



Source: JRC based on PRODCOM data

In 2024, the unitary production value was at around EUR 0.10 per watt, a 44% decrease from EUR 0.17 per watt in 2023 and below the average unitary production value (EUR 0.16) for the past decade (2015-2024) (Figure 26).

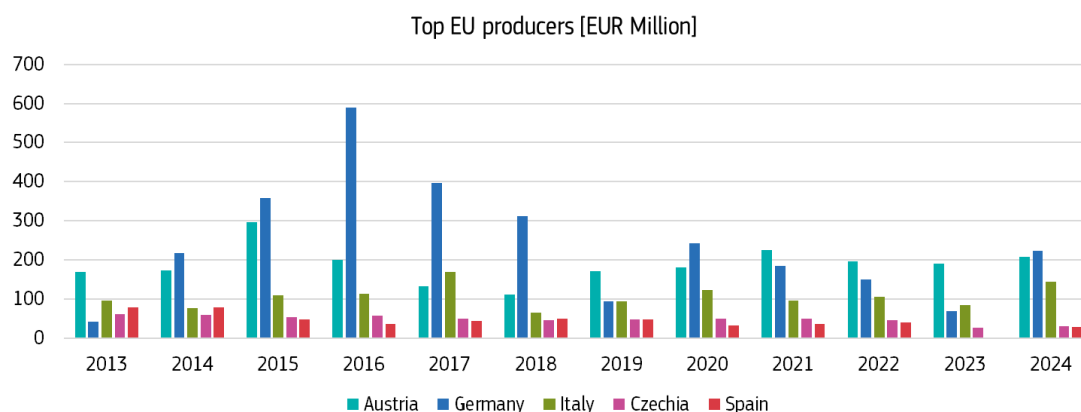
Figure 26. EU production value per quantity of hydraulic turbines [EUR per Watt]



Source: JRC based on PRODCOM data

Over 2012-2023, Austria, Germany, Italy, Czechia and Spain were the top EU producers, but not all Member States had disclosed data for all the years (Figure 27). Germany and Austria combined hold more than half of the EU production. It must be noted that Production values refer exclusively to turbines and associated parts. If all the technologies associated to the hydropower chain were considered (approx. 150 types, e.g. including pipes, penstocks, valves, gates), the European House Ambrosetti found that these numbers would be EUR 27.7 Billion and EUR 43.4 Billion for Italy and Germany, instead of EUR 107 Million and EUR 150 Million in 2022, respectively, for a total EU's Production value of EUR 132.3 Billion from the first ten EU member states, instead of EUR 562 Million in 2022 ⁸⁵.

Figure 27. Top EU producers among the Member States disclosing data [EUR Million]. Source: JRC based on PRODCOM data.



Source: JRC

4 EU market position and global competitiveness

4.1. Global markets and growth prospects

4.1.1 Current EU positioning in the global market

European companies are global leaders. The large European operators (EDF, EDP, ENEL, ENGIE, ENBW, Iberdrola, PPC, Statkraft, UNIPER, Vattenfall, amongst others) continue to invest in many hydropower projects outside of Europe. Many European engineering and consultancy companies offer knowledge, expertise, or consulting to hydropower projects outside of Europe, where there is considerable growth in the hydropower sector (Artelia, Lombardi, ISL, AFRY -former Pöyry and AF-, Sweco, MESYSolexperts, Tractebel -former Lahmeyer and Coyne et Bellier-, amongst others). Meanwhile, many construction companies (Webuild, Skanska, Strabag, Vinci, Walo, amongst others) act as civil contractors or even as EPCs (Engineering, Procurement and Construction) in the framework of turnkey projects⁶. EU manufacturers hold 39% of extra-EU global exports, as well as a very large share of high-value inventions.

To support European hydropower, the European Hydropower Alliance was recently founded, and it gathers the following energy utilities: EDF, EDP, Enel, Engie, Fortum, Iberdrola, Statkraft, Uniper, Vattenfall and Verbund. The Alliance is committed to advancing sustainable hydropower's role in achieving the EU Green Deal and Fit for 55 ambitions. The Alliance supports the development of an integrated decarbonized electricity market in Europe by providing affordable, renewable, dispatchable, and secure electricity which ensures the maximum integration of renewable sources of energy.

The Alliance's goal is to ramp-up the visibility of hydropower as:

- A vital source of energy and flexibility contributing to EU security of supply and energy independence.
- A key renewable energy source with development potential contributing to decarbonisation, adaptation to climate change and resilience.
- A sustainable energy source respecting biodiversity.
- A generation technology with multi-purpose non-power related benefits.

The major hydro turbine manufacturers in the hydro turbine market include General Electric Co, Andritz AG, Voith GmbH & Co KGaA, Dongfang Electric Corporation Ltd, Harbin Electric Corp, Power Machines, Toshiba Corp, and Hitachi Mitsubishi Hydro Corp. These hydro turbine companies together hold more than a 75% share of the cumulative installation in the global hydro turbine market with the remaining shared by local and other international players. The first three (EU) companies shared 50% of the market in 2021⁸⁶.

4.1.2 Market prospects

In the EU, as explained before, the potential for new large hydropower systems is not very high, except for new closed-loop PSH, for which several GW are planned or under construction. The EU's, hydropower fleet needs to be refurbished and modernized. This would allow to operate it also as system for climate change adaptation (water storage), also supported by the EU Water Resilience Strategy, and for better integration of the volatile renewable sources into the grid. Innovative small hydropower systems will be installed in existing water and wastewater systems, also supported by policy contexts like the energy neutrality targets of the recast Urban Wastewater Treatment Directive or the Social Climate Fund. In the last decade, 12 GW are estimated to have been refurbished.

The hydropower market is instead growing elsewhere. For small hydropower (SHP), Precedence Research (2024) estimated that the global SHP market size accounted for USD 2.26 Billion in 2024 and it is estimated to surpass around USD 2.99 Billion by 2034, growing at a CAGR (Compound Annual Growth Rate) of 2.82% from 2024 to 2034. 58% is shared by mini hydropower (up to 1 MW). Europe, as a whole, shares 28% of this market. According to Precedence Research (2024), the Asia Pacific small hydropower market size is worth around USD 840 Million in 2024 and is growing at a CAGR of 2.98%⁸⁷. Therefore, increasing knowledge and technology export of European companies to these countries represents a great opportunity. Within this context, the EU's Horizon research and innovation programme projects HYDRO4U and HYPOSO aim to improve the EU export capacity to foreign countries. Hydro4U aims at demonstrating European small hydropower equipment and technologies in Central Asia. Eco-friendly hydropower solutions will be implemented at two demonstration sites. HYPOSO aimed at supporting the EU's industry to foster more clean and reliable energy in target countries in Africa and Latin America.

Concerning large hydropower, the large EU companies working in civil construction, and the largest EU equipment manufacturers, could continue to lead the extra-EU export market with their high quality and innovative manufacturing equipment. There is an increasing interest towards hydropower also in the US, where EU companies already export a lot. However, such a leadership is undermined by several factors, better discussed below. The EU share on global extra-EU exports is decreasing year by year.

4.1.3 General trends and POTEnCIA's CETO 2025 Scenario

Hydropower is currently the giant of low-carbon electricity technologies and it is the key technology for an optimal integration of volatile energy sources (e.g. wind energy and photovoltaics) into the electric grid. Hydropower (together with water & energy storage) capacity needs to increase in the near future, but it faces several environmental constraints and barriers. The International Energy Agency (IEA) and the International Renewable Energy Agency (IRENA) estimate a need for around double the amount of hydropower that is currently installed in their net zero scenarios for 2050. IRENA's 2050 net zero scenario has more than 2,900GW total hydropower capacity, of which almost 420 GW is PSH. Global hydropower installed capacity is expected to expand from 1416 GW in 2023¹⁴ to about 1780 GW in 2030 and up to 2156 GW in 2050 according to the *Global CETO 2°C scenario 2025*².

In the EU, the outlook for the expansion of hydropower is less bright, due to several barriers that are extensively discussed throughout the report and in Table 1⁸⁸. The main technical priorities for hydropower in the EU are defined in the EU Clean Energy Transition Partnership⁸⁹, and are flexibility, storage, digitalisation, services of hydropower, sustainable solutions and sediment handling. An urgent need is to support the EU manufacturing capacity to keep its leadership position on the global export market. The most important administrative challenges are the following ones: 1) the regulatory context requires stringent environmental, climate and wildlife standards that entail administrative hurdles; 2) the most suitable locations for hydropower projects have already been exploited or are environmentally protected areas⁹⁰; 3) new large greenfield hydropower plants face high investment costs and/or long construction times that expose them to risk and to complex financing processes; 4) remuneration of hydropower plants especially in the long-term is challenged by the expected reduction of electricity prices, especially due to the penetration of more renewable energy. Ad-hoc policy interventions and robust implementation of the Electricity Market Design (EMD) reform are needed to tackle these barriers that undermine the hydropower fleet in the EU, which is so critical to the accomplishment of the EU carbon neutrality objective.

According to *POTEnCIA's CETO 2025 Scenario*⁹¹ (see Annex 5 for more details on the POTEnCIA model), hydropower installed capacity in the EU is assumed to grow with +8.3 GW of new PSH in

² A short description of the *Global CETO 2°C scenario 2025* and the POLES-JRC model can be found in Annex 5.

2050, see Figure 28, +0.45 GW of reservoir hydropower and +0.97 GW of run-of-river hydropower. Closed-loop PSH generation (plus that from the pumped water in open-loop PSH) will produce an additional +67 TWh in 2050 with respect to 2025; RoR will produce + 4 TWh/y and RSHP (also called dam hydro in POTEnCIA) generation is projected to increase by 2 TWh in 2050. These analyses are based primarily on techno-economic considerations and the surrounding political framework. However, they do not consider future water availability, because of the high uncertainty and complexity of projecting the future impact of climate change. Figure 29 and Figure 30 show the new installation trend and global projections, respectively.

Figure 28. Hydropower installed capacity and electricity generation in the EU, 2025-2050. PSH generation includes closed-loop PSH and that from the pumped water in open-loop PSH.

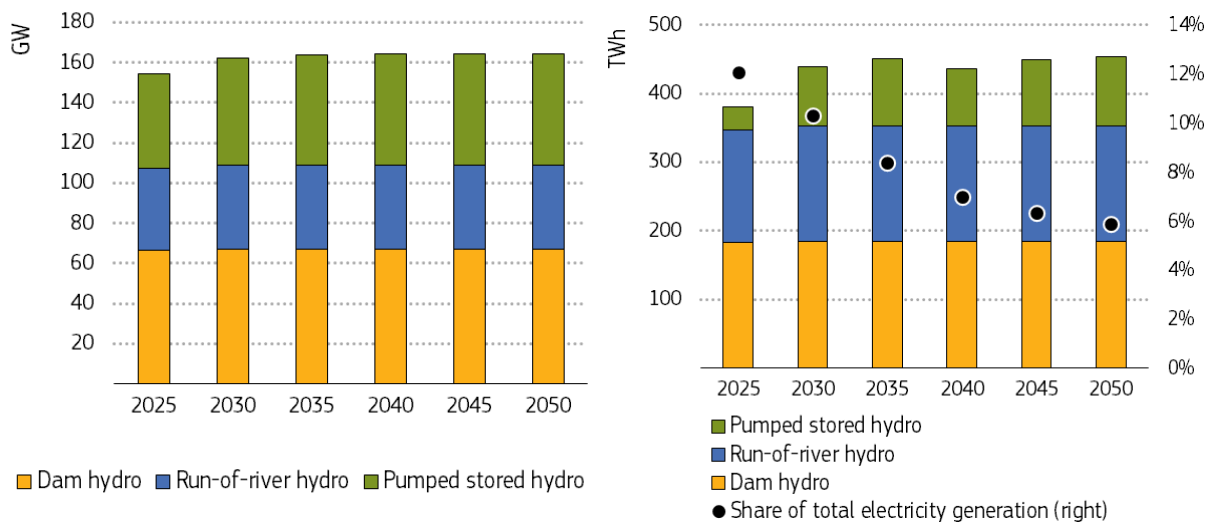


Figure 29. New hydropower installations over the years.

Hydro - Annual new installations - 5 year average: EU (left) and global (right)

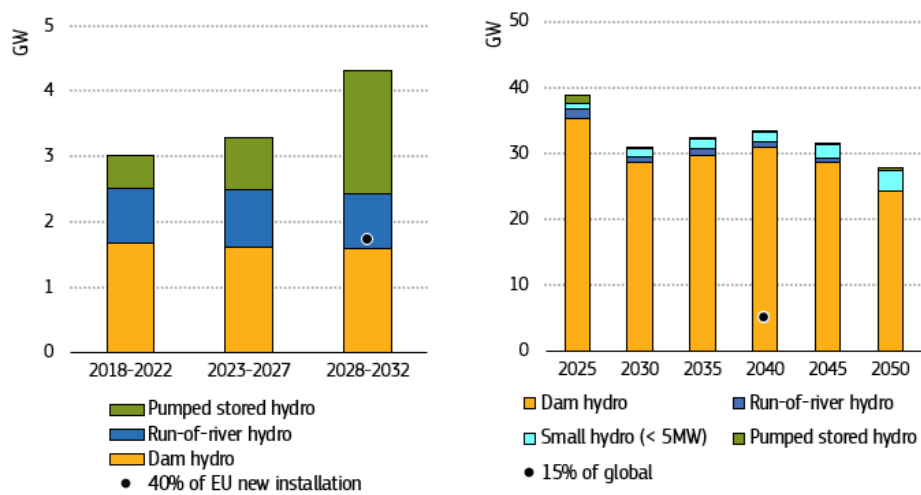
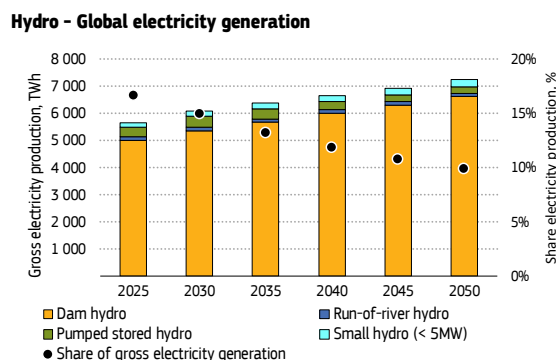


Figure 30. Global trend on hydropower.



Source of figures 5-6-7: POTEnCIA CETO 2025 Scenario (see Annex 4.1) and Global CETO 2°C scenario 2025 (see Annex 4.2) for the global trend.

Tarvydas (2022)⁹² projected an increase of hydropower generation up to about 420 TWh/y in 2050 (including PSH), thus an increase of about 60 TWh/y with respect to the average annual generation recorded in the last 10 years (+17%). By further elaborating results of Calheiros et al. (2024)⁹³, by 2040 traditional hydropower generation (TWh/y) may increase by 27%, by 2060 by 16%, while predictions are highly uncertain beyond 2060. According to Schmitt and Rosa (2024)⁹⁴, who elaborated data of IEA, in OECD Europe the demand for hydropower installed capacity (GW) may increase by 30% by 2050.

4.2 Trade (Import/export) and trade balance

For monitoring trade in hydropower section, the selection of trading goods aligns with the Net-Zero Industry Act (NZIA) list of components^{aa}. International trade is monitored using six-digit codes of the Harmonised System (HS) classification⁹⁵ dedicated to hydraulic turbines and their parts (e.g. when traded separately) (Table 11), although these components represent only a small fraction of the total hydropower equipment, as discussed for the Production value.

Global exports rose from EUR 791 Million in 2023 to EUR 932 Million in 2024, with EU exports (including intra-EU trade) increasing from EUR 351 Million to EUR 452 Million. Over 2022–2024, the EU accounted for 47% (49% over 2020–2022) of global exports (including intra-EU trade), while extra-EU exports (excluding intra-EU trade) represented 38% of global transactions (42% over 2021–2023). The EU met 80% of its import needs through intra-EU trade (Figure 31 and Figure 32). Extra-EU exports include exports from EU companies to companies based outside of the EU, thus excluding intra-EU market.

Overall, the EU has maintained a positive trade balance between 2014 and 2024, averaging approximately at EUR 302 Million. In 2024, extra-EU exports increased by 18% to EUR 291 Million, while extra-EU imports decreased by 22% to EUR 25 Million, yielding a trade surplus of EUR 266 Million. This surplus was below the 11-year average but marked a 24% improvement compared to 2023.

^{aa} C(2025) 9034 final, https://single-market-economy.ec.europa.eu/document/download/860c257b-16db-429e-91c1-a30bea57e7ce_en?filename=C_2025_9034_1_EN_ACT_part1_v8.pdf

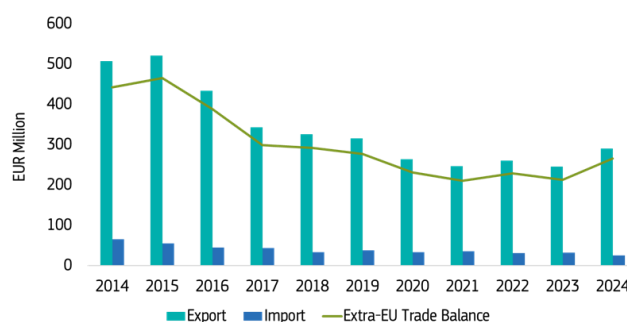
Notably, both extra-EU imports and exports are predominantly composed of parts of hydraulic turbines (HS 841090), followed by large turbines (HS 841013) (Figure 33).

Table 11. Selection of HS codes as a proxy for monitoring trade for hydropower.

HS code	Description	Alias
841011	Hydraulic turbines and water wheels, of a power ≤ 1.000 kW (excl. hydraulic power engines and motors of heading 8412)	turbines S
841012	Hydraulic turbines and water wheels, of a power > 1.000 kW but ≤ 10.000 kW (excl. hydraulic power engines and motors of heading 8412)	turbines M
841013	Hydraulic turbines and water wheels, of a power > 10.000 kW (excl. hydraulic power engines and motors of heading 8412)	turbines L
841090	Parts of hydraulic turbines and water wheels incl. regulators	parts

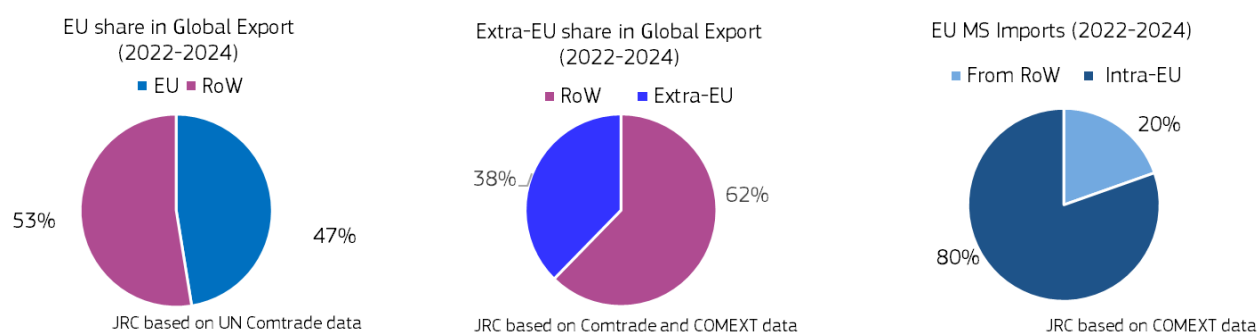
Source: Harmonised System

Figure 31. Extra-EU trade for hydraulic turbines and parts [EUR Million]



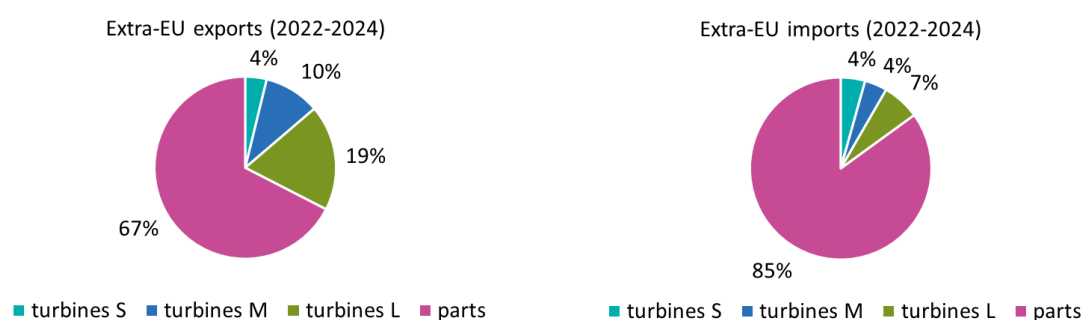
Source: JRC based on COMEXT data

Figure 32. EU share in global export (left), extra-EU share in global export (middle) and EU imports (right) [2022-2024].



Source: JRC based on COMEXT and COMTRADE data

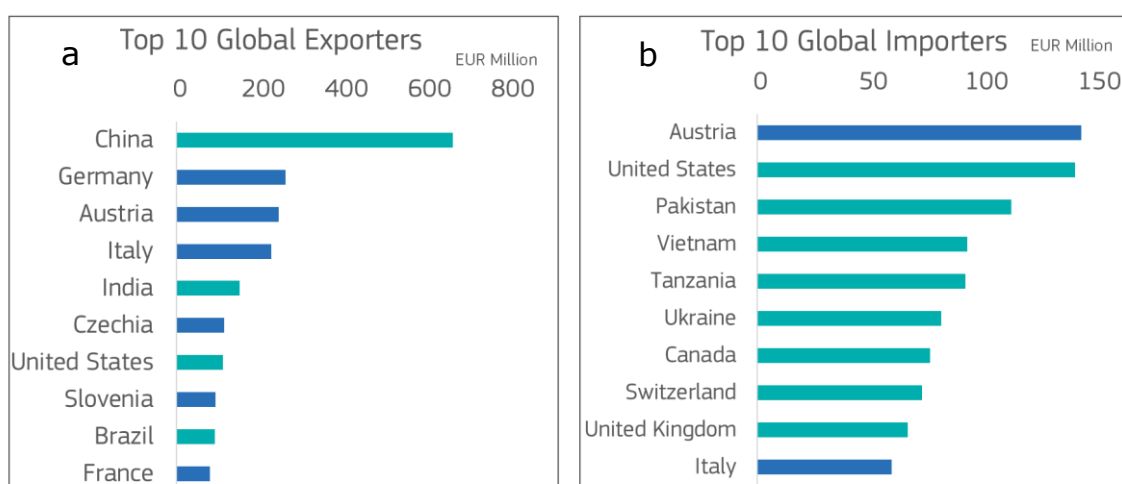
Figure 33. Traded value share of hydropower [2022-2024].



Source: JRC based on COMEXT data

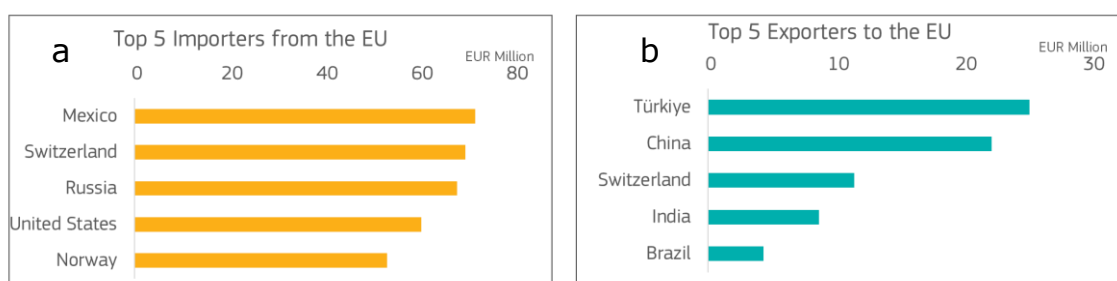
China was the leading global exporter, accounting for 26% of global exports, followed by Germany (10%) and Austria (10%). Austria and the US were the largest global importers (Figure 34), but for Austria these were mainly intra-EU imports.

Figure 34. Top ten global (a)exporters and (b)importers for 2022-2024 [EUR Million].



Source: JRC based on COMTRADE data, 2025.

Figure 35. Top five countries (a) importing from and (b) exporting to the EU for 2022-2024 [EUR Million].



Source: JRC based on COMEXT data, 2025.

During the same period (2022-2024), Mexico and Switzerland were the largest importers from the EU, receiving 9% of the extra-EU exports each, followed by Russia (8%). Türkiye was the largest exporter to the EU, accounting for 28% of extra-EU imports, followed by China (25%) and Switzerland (13%) (Figure 35).

Table 12 shows the growing markets^{bb} of hydraulic turbines and parts during 2021-2023^{cc}. Tanzania recorded the largest net import increase, followed by Vietnam and Australia, where the EU captured 1%, 16% and 16% of each country's growing market, respectively. The EU also secured expanding markets in the Switzerland (96%), Philippines (61%), United Arab Emirates (44%), the UK (43%) and the US (32%).

Table 12. Growing markets of hydraulic turbines and parts based on a 2-year average of net import change.

Country	Total import (2021-2023) [EUR Million]	% import from the EU	2-year average of net import change
Tanzania	99	1%	56
Vietnam	123	16%	22
Australia	32	16%	22
Indonesia	61	16%	19
United States	107	32%	15
United Kingdom	55	43%	11
United Arab Emirates	40	44%	6
Laos	33	2%	4
Switzerland	64	96%	2
Philippines	33	61%	1

Source: JRC based on COMTRADE data

For the future, a great opportunity for EU companies is their export potential associated to innovative equipment and small hydropower (high-value inventions), as well as assistance in the overall design and operation of hydropower plants.

4.3 Status of net zero technology systems and components in the EU

The results presented in this section mainly rely on the further elaboration of data previously presented. Furthermore, data gathered by a survey among EU manufacturers is used to shed light and reflections on EU manufacturing capacity, its potential expansion and related challenges. It is worth to say that turbine manufacturing is by no means a mass manufactured product, rather each turbine is tailored (designed and manufactured) for a specific installation. Each hydro plant is unique and has the potential to generate new knowledge.

^{bb} Calculated as $net\ import\ change = [(import_{2022} - import_{2021}) + (import_{2023} - import_{2022})]/2$

^{cc} Latest year data (2024) is incomplete for comtrade, because it does not provide estimates for the missing values as comext does.

4.3.1 Relevant final products and primarily used components

A hydropower plant is composed of civil and electric, hydraulic and mechanical components. Considering hydromechanical components, the following components have been identified as primarily used:

- Hydro turbine runners
- Distributor with adjustable / guide vanes
- Large hydro butterfly valves
- Large hydrospherical valves
- Large hydro hollow-jet discharge valves

The first two are also included in the main specific components. ETIP Hydropower and other manufacturers were also consulted (see Acknowledgement section).

4.3.2 EU Manufacturing Benchmark

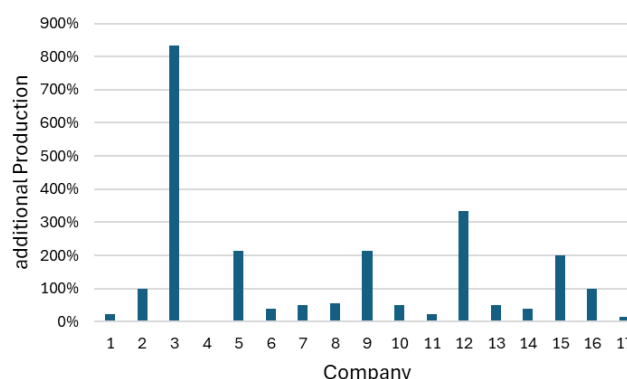
By combining Eurostat data on installed capacity (MW), and CETO data on Production values (MW) of turbines and parts and extra-EU exports (EUR, and with the average conversion factor EUR/MW from the Production value as first approximation) of turbines and parts, in the last 10 years, 16.6 GW remained in the EU, but new installed capacity only increased by 5 GW, that means that 11.6 GW might have been refurbished within the EU.

Considering the projection of POTEnCIA CETO 2025 Scenario of +8.3 GW of PSH in 2050 from 2025, +0.45 GW of reservoir hydropower and 0.965 GW of RoR hydropower, the need of new hydro turbines is + 9.7 GW over 15 years, i.e. an annual deployment need of 0.65 GW/year. The annual production in 2024 was 4.2 GW (an average of 2.8 GW in the last decade), which, according to a recent survey across the EU (see figure below), is below the manufacturing capacity. Therefore, assuming the production as a lower proxy for manufacturing capacity, the NZIA benchmark of at least 40% of the Union's annual deployment need in 2030 can be achieved.

Figure 36 shows the planned increase of manufacturing capacity that some respondents have estimated and communicated in an EU-survey. EU manufacturers clearly showed that their manufacturing capacity is higher than the one requested in 2024. Considering that 2024 was already a very positive year for the production of turbines in the EU, this means that the EU hosts a large potential for export, but that might not be very competitive in the future on the global market due to the high labour costs. Therefore, the maximum achievable production is the one in 2024 plus the increase depicted in Figure 36. The largest additional capacity (800%) is from a company that manufactures pipes and penstocks for hydropower.

Although new large manufacturing facilities seem not to be planned in the EU as general trend, some EU companies have built new facilities outside of the EU to better serve the markets in those countries, with investments of tens of million EU and payback times estimated to be a few years, or below. The largest challenge is to keep the global leadership in manufacturing of hydro turbines that EU companies have been holding for a century, and that are gradually losing due to global competition and lack of support on manufacturing capacity, especially on reducing costs.

Figure 36. Additional manufacturing capacity of EU hydropower manufacturing companies at their existing facilities, with respect to actual production in 2024.



Source: own survey.

4.3.3 EU share in Global Manufacturing Benchmark

The EU has a long history in hydropower manufacturing. The three largest EU companies play a pivotal role on the global market, and, together with the largest turbine manufacturers for small hydro, they share up to 50% of the global market. Precedence Research (2024) estimated that the global SHP market size accounted for USD 2.26 Billion in 2024 and it is estimated to surpass USD 2.99 Billion by 2034, growing at a CAGR (Compound Annual Growth Rate) of 2.82% from 2024 to 2034. 58% is shared by mini hydropower (up to 1 MW). Europe, as a whole, occupies 28% of this market (Precedent Research, 2024). According to data presented in the previous section, trade is positive with a large export outside of the EU from EU companies, and with the largest share of imports relying on intra-EU imports. The hydropower sector does not significantly rely on critical materials (see last section). Considering the centenary history of the EU hydropower, that the three largest global hydro manufacturers are in the EU and that in the EU hydropower is a well affirmed sector it is possible to state that most of components in the EU hydropower fleet is EU made (also stated in ESHA, 2001). However, some challenges hamper EU competitiveness on the global market.

One of the biggest issues for turbine manufacturers is to hire skilled and smart workers in manufacturing and field service. Moreover, in order to face competition it is today common practice among the EU well known manufacturing companies to use subcontractors to a much greater extent, becoming almost only an assembly workshop. This reduces the risk in periods of scarcity of orders. Nevertheless, giving out the work to small workshops lacking the knowledge of turbine design and function makes it almost impossible to maintain the quality leadership at the world level. Furthermore, some developers try to reduce initial costs to get investments to correspond to the decreasing electricity prices. This means importing simple and cheap equipment from non-EU manufacturers. Additional risks are also represented by China competition on the global market, as well as an emerging competition from US companies which are receiving a huge support from the government to increase their manufacturing capacity, especially on additive manufacturing, which will undermine the penetration of EU companies in the US hydropower market.

4.4. Resources efficiency and dependence in relation to EU competitiveness

Hydropower and pumped-storage hydropower are of strategic importance to the EU energy system and can contribute to the EU resilience and independence⁹⁶.

The hydraulic and mechanical equipment of hydropower is typically made of materials that are

available in most parts of the world, such as steel, concrete, and – to a lesser extent – copper, so that hydropower expansion may not be limited by material availability. Concrete is used for dam construction and the required civil works, including the power station. In large-scale stations, concrete is used in the construction of tunnels and caverns. The manufacture of mechanical components typically uses steel. The steel used in the turbines and concrete in buildings contribute significantly to the overall impact of the hydropower plant. Local materials are typically used, and this explains the high added value of hydropower to local economies. Copper is used at relatively low quantities in the generator sets. Over the last decade, novel materials have been introduced in the hydropower sector and/or are under testing, e.g. fibre-reinforced composites for small-scale hydropower⁶³. It is estimated that the total weight of the electromechanical equipment (runner, distributor, generator, draft tube and casing) amounts to almost 900 ktons in the EU (Quaranta, 2023)⁹⁷.

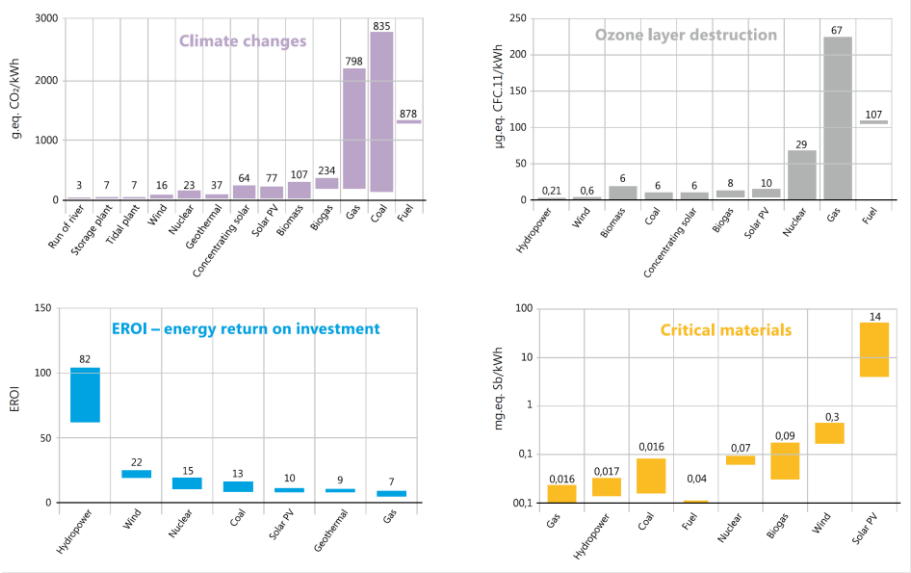
Hydropower equipment does not contain critical materials such as lithium and cobalt (used in electric vehicles), or neodymium, praseodymium, and dysprosium (used in electric vehicles and wind power plants). Therefore, hydropower is the best renewable energy for reducing pressure on mineral resources (Figure 37 and Figure 38). The Extraction of Mineral Resources indicator is measured in kilograms of antimony equivalent (kgeq.Sb) per kWh generated: the value for hydropower is 0.017, while it is 0.04 for coal, 0.3 for wind and 14 for solar PV⁹⁸. Figure 37 summarizes these findings and more specific details can be found in the IEA (2023)⁹⁹ report, which highlights the wide use of critical material in the other renewable energy technologies. This is also confirmed in Andronova et al. (2022)¹⁰⁰. However, the values calculated on PV reflect the impact of PV energy of years ago^{dd}, and since that time the performance has improved a lot¹⁰¹.

Variable-speed machines are based on two emerging generator types that use power electronic converters: the doubly fed induction machine and converter-fed synchronous machine (Kougias et al. 2019). Variable-speed permanent magnet generators could be an alternative for very low-head hydropower (water wheels, Archimedes screws). However, the material components of permanent magnets may suffer from shortages in the near future, worsened by the Chinese supply monopoly (the EU plays a significant role only in the assembly stage, where its share is above 50%)^{102, 103}. This should stimulate the development of novel electro-mechanical equipment and the improvement of the lifecycle of such materials (e.g. recycling). Finally, hydropower development may involve substantial excavation and tunnelling, requiring significant amounts of energy to run the appropriate machinery, and concrete to build the dam. Especially for the countries where hydropower is expanding significantly, the construction of large dams can require large amount of concrete, which generates large carbon emissions. Furthermore, concrete depends on sand availability, which is the second most used resource on Earth, after water. Sand is often dredged from rivers, dug up along coastlines and mined. This aspect should be taken into consideration: the International Commission on Large Dams (ICOLD) is looking at this aspect, and it is aiming at reducing the carbon footprint of dam construction, for example by using new construction methods and materials (ICOLD pers. comm., HYDRO2023 conference, and see e.g. Wolfsborg et al., 2023)¹⁰⁴.

Threats and opportunities within the current social, energy, geopolitical and climate situation are discussed in the [2022 CETO report](#).

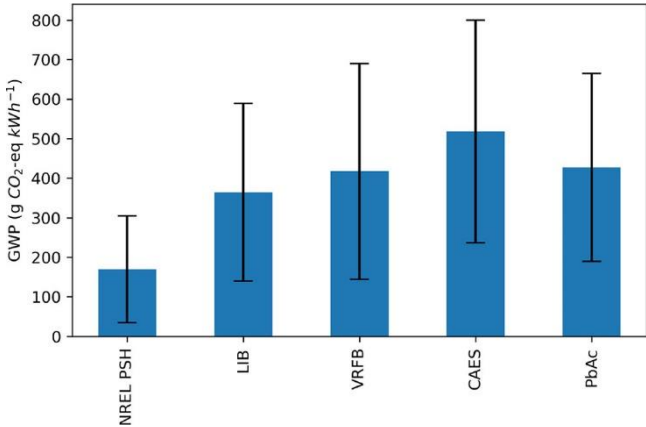
^{dd} Note that the critical mineral issue is too complex to be resumed to a single value. Furthermore, the characterization factor used to aggregate the values in Sb.eq also relies on USGS (U.S. Geological Survey) extraction data from the last century, which clearly has some limits. For the mineral impact of the PV value, if it relies on theecoinvent value, most of the impact here presented is due to indium that is considered as depleted when zinc is extracted from mines (so it is more related to the way some LCI are built than to the use of a really critical material for PV).

Figure 37. Impact indicators from ETIP Hydropower Europe (EU project), with minimum and maximum values (the bar) and the average value (the numbers). EROI, also sometimes called the energy returned on energy invested (EROI) or recovery factor of energy or gain factor of energy and is the ratio of the amount of usable energy (the energy) delivered from a particular energy resource to the amount of energy used to obtain that energy resource.



Source: ETIP Hydropower

Figure 38. Global warming potential (GWP) of closed-loop PSH (NREL PSH), compared to literature GWP values for lithium-ion battery storage (LIB), vanadium redox flow batteries (VRFB), compressed-air energy storage (CAES), and lead-acid battery energy storage (PbAc). The bar heights indicate the mean GWP for each technology, and the error bars indicate the GWP standard deviation¹⁰⁵.



Source: NREL

5 Conclusions

Hydropower is the largest global renewable energy source to date. Hydropower provided, on average, 360 TWh/y in the EU in the last decade (including PSH, that adsorbed and generated approx. 40 TWh/y). A quarter of the global PSH installed turbine capacity is in the EU.

The hydropower sector is characterized by several strengths and advantages with respect to the other renewable and storage technologies. Reservoirs provide additional services in emergency situations, e.g. water and energy storage for irrigation and fire-fighting, flood control and drought mitigation, and navigation. On the other hand, these uses may sometimes be in conflict one another or not recognized. The Energy Return on Investment (EROI) is the highest among the renewable energy sources, as well as the life-cycle emissions. This is mainly due to the fact that traditional hydropower equipment does not use rare and critical materials, and capacity factors and efficiencies are generally higher than those of photovoltaics and of wind farms. The life span of hydropower is also longer. Hydropower is the most flexible technology, providing flexibility and stability services to the grid, by for example preventing, and re-activating the grid after, blackouts. As the penetration of variable energy sources (mainly wind and solar power) increases, the flexibility provided by hydropower operation is essential. PSH is a mature technology and, as a result, its technological and manufacturing/market position is considerably more advanced than that of other energy storage technologies (e.g. battery storage, flywheel, thermal and chemical storage). PSH can store water-energy (with daily, monthly and seasonal storage depending on the installed capacity and reservoir's volume) more cost-effectively (the lowest Levelized Cost of Electricity – LCOE-) than any other option and can put and absorb energy available in seconds.

However, several obstacles and challenges exist. The first major barrier is the effort to simultaneously pursue energy, climate and environmental goals, where hydropower exhibits diverse and sometimes controversial roles. Depending on the context, hydropower can generate several adverse effects on the environment, although it is shown that river fragmentation goes well beyond hydropower barriers. The most suitable sites in the EU have been already exploited or are protected by environmental legislation (e.g. protected areas, natural parks), so that new large plants for traditional hydropower generation would encounter higher costs and longer development time frames. Hydropower projects have longer pre-development, construction and operational timelines than other renewable energy technologies and are generally characterised by high investment costs, hence have higher financial risk, requiring specific policy instruments and incentives as well as a longer-term policy perspective and vision. A cost-effective way to ensure secure and affordable energy supplies is a well-functioning and integrated EU's energy market. However, European hydropower operators are not remunerated for all their services (e.g., flexibility). Therefore, another major challenge is defining a value for all the benefits, which is necessary to allow discussions and negotiations between different water users. Public sector involvement is critical for hydropower, and innovative financial mechanisms are crucial for equitable and efficient sharing of benefits among water users. The challenge is to find ways of framing long-term strategies, securing long-term finance sources, and shielding them as effectively as possible from short-term constraints. The hydropower growth outside of the EU is evident, but EU companies are struggling in keeping their leadership position.

The expansion capacity of green-field large hydropower in the EU is smaller than for the other renewables. According to the existing scientific assessments, the current hydroelectricity generation could be incremented by 10% by lightly modernising existing hydropower (partly offsetting climate change effects and limitations imposed by environmental legislation). An additional +3% can be achieved by exploiting the hidden potential in existing infrastructures (mills, existing barriers, wastewater treatment plants and water distribution networks). Small-scale hydropower opportunities in rural contexts and integrated in existing facilities can provide decentralized energy when the electric grid is not available, difficult to be connected or to avoid further expansion of the grid. New water storage systems, designed as multi-purpose projects, may be required for water

management services that could be exploited also for energy storage and generation. New reservoirs may be useful to mitigate the already visible effects of climate change (floods, droughts and fire), and in the Alps to store additional water from glacier retreat. Capacity expansion of PSH is of some tens of GW, including using abandoned mines and exploiting existing systems.

Hydropower exhibits high hybridization potential. Photovoltaic (PV) systems can be installed as floating solution on hydropower reservoirs to reduce PV land use and optimize the overall efficiency of the hydropower-PV power plant. Due to the different characteristics in response time and storage volume, hydropower and batteries can make a perfect combination in many cases. Waste-heat can be extracted from the cooling system of the generator. Hydropower technologies can be used, after adaptation and optimization, for tidal and wave power generation.

Hydropower companies within the EU are very competitive and are of strategic relevance, especially considering the current geopolitical situation. European companies own a great export capacity of their products and knowledge and new turbine technologies. EU countries hold 39% of the extra-EU global exports. Data of production presented in this report only focuses on turbines and their parts, but the overall dimension of the market (pipes, civil works) may be 2-orders of magnitude higher, whose data are difficult to track. EU is leading supplier and all targets from NZIA legislation are already achieved. EU companies are also very acknowledged for their experience in the fields of sustainable and mitigation solutions and in the O&M, exerting their consulting services worldwide. 16% of the high-value inventions of the top 10 companies is shared by two EU companies. Furthermore, the EU is well positioned in terms of scientific publications, with the main competition of China.

Therefore, to keep a competitive EU hydropower sector in an increasingly challenging world, the strong competence (scientific and industrial) of EU's companies and institutions is key. This is also essential to support the renewable energy targets in the EU. It is essential to increase public awareness about the benefits of hydropower, as a required catalyst for a safe and independent energy transition. Information availability, dialogue with society, and strategies towards social acceptance of associate benefits in different social, technical, environmental and economic contexts are actions that require immediate consideration. At the same time, hydropower developers must consider the complex effects on the environment. A scientifically based process to find a balance between different policy targets, impacts and benefits, and conflicting interests of multi-purpose reservoirs, are essential.

Hydropower projects are inherently multidimensional, generating technical, economic and environmental knowledge while posing complex challenges. Hydropower "made in the EU" strengths are often missing in some other renewable technologies. Hydropower role for the EU goes beyond its future expansion capacity: hydropower current capacity plays a catalyst role for the energy transition and, when reservoirs are designed with an integrated approach, it can also provide water resources management opportunities. Strengthening collaboration among policymakers, scientists and industry is essential for strategic decision-making.

Annex 1: Sustainability assessment framework

Table 13. Summary table with sustainability indicators, improved from the 2024 one.

Parameter/Indicator	Input
Environmental	
<i>LCA standards, PEFCR or best practice, LCI databases</i>	Yes ¹⁰⁶
<i>GHG emissions</i>	The IPCC states that the global hydropower generates a median greenhouse gas (GHG) emission intensity of 24 gCO ₂ -eq/kWh (grams of carbon dioxide equivalent per kilowatt-hour of electricity generated allocated over its lifecycle). By comparison, the median figure for gas is 490 gCO ₂ -eq/kWh. However, it depends on the geographic context, e.g. alpine or non-alpine area, and the regional context, as GHG emissions in the European context, generally involving reservoirs in alpine contexts, are lower than in tropical contexts. Hydropower helps in reducing CO ₂ emissions ¹⁰⁷ . Additional considerations can be drawn from Gemechu and Kumar (2022) ¹⁰⁸ , who reviewed the gCO ₂ eq/kWh associated to several hydropower plants (HPPs), from their manufacturing to their dismantling. Below 5 MW, results are quite dispersed because they concern micro/mini hydropower plants. Indeed, the relative impacts of such HPPs are very low when installed in existing structures and in artificial existing canals (and still lower than those of larger HPPs), while relative impacts can be quite high when associated to new barriers in freshwater systems. Their relative impact ranges between almost 0 and 75 gCO ₂ eq/kWh (although most of them range between 0 and 35 gCO ₂ eq/kWh). Above 5 MW, the relative emissions are relatively small and quite constant, generally below 5 gCO ₂ eq/kWh, and may be mainly associated to the reservoir. If European case studies are only used, no correlation between emissions and installed power was found.
<i>Energy balance</i>	EROI = 60-100 ¹⁰⁹ , the highest one amongst energy technologies. Water reservoirs provide storage capacity and hydropower plants provide flexibility. EROI (Energy return on investment) is a ratio that measures the amount of usable energy delivered from an energy source versus the amount of energy used to get that energy resource
<i>Ecosystem and biodiversity impact</i>	New barriers can generate several impacts. Obsolete and existing barriers are also the cause of river fragmentation, but most of them are not for hydropower or may not have hydropower as primary use.
<i>Water use</i>	The average flow rate discharged by European reservoir-hydropower plants is 10.5 m ³ /kWh (elaborating data of the JRC hydropower database). Water is used for energy generation and released downstream in ROR plants, while the natural hydrological regime is altered, and some evaporation losses occur, in reservoirs plants (RSHP). The water footprint of hydropower in EU during construction phase is 3.6 m ³ /GWh, which is 90-fold less than the solar one (in 2019, but now PV technology is improved and this difference may be slightly attenuated) and similar to the wind one ¹¹⁰ . The water footprint for operation (excluding the turbined flow) is almost zero for ROR and 32 m ³ /MWh for RSHP (evaporation mainly), more than 1-order of magnitude higher than that for PV, that is 0.32% of the discharged flow (assuming 10.5 m ³ /kWh abovementioned). Globally, around 507 GW of hydropower competes with irrigation. While hydropower reservoirs might support irrigation, there are well-known cases where it reduces water availability for irrigated food production, e.g. in Portugal ¹¹¹ . On average in the EU, the storage volume of reservoir is 12% of the annual inflow volume to the reservoir.
<i>Air quality</i>	No direct pollutant emission during operation, but methane emissions might occur from some reservoirs due to organic matter decomposition ¹¹² . Nevertheless, this is not a serious issue for the EU.
<i>Land use</i>	Hydropower densities (W/m ²) vary quite widely across the literature as they are dependent on geographic and topological conditions. Reservoirs include the additional land required for impoundment, but that may also serve for multiple purposes. Elaborating data of ICOLD 2023 database, the density value for reservoir hydropower is 34 W/m ² (50° percentile) in single purpose reservoirs, while it is 10 W/m ² and 2 W/m ² in multi-purpose reservoirs where hydropower is the first or second (or more) use, respectively, considering the reservoir area as water footprint. If the 50° percentile value is multiplied by the capacity factor (0.38), by the power at operating point (90% of the installed one) and by the infrastructure ratio (0.62), the power density becomes approx. 6 W/m ² , higher than the average global value ranging from 0.25 to 0.75 W/m ² calculated in van Zalk and Behrens (2018) ¹¹³ . The infrastructure ratio (that represents the additional surface for mines, roads, foundations, etc.) is 0.62, higher than wind (0.10)

	and slightly lower than photovoltaics (0.73-0.91). The infrastructure ratio of biomass power plants is the highest (0.90) ¹¹³ (infrastructure ratio =1 is the optimal/ideal value). The energy density value of wind farms is comparable with that of hydropower plants ¹¹⁴ .
<i>Soil health</i>	Alteration of sediment transport, possible landslides in big projects, but also improved maintenance and additional monitoring of the environment.
Hazardous materials	Critical materials are not commonly used.
Economic	
Parameter/Indicator	Input
<i>LCC standards or best practices</i>	Zakery and Syri (2015) ¹¹⁵ found that the annualized life cycle cost (LCC) for PSH ranged between 200 and 270 EUR/kW/y, half the LCC of batteries. Donnelly et al., (2010) ¹¹⁶ found that the LCC was 66 \$/MWh for a reservoir hydropower plant, 88 \$/MWh for a ROR plant, and 103 \$/MWh, 405 \$/MWh, 99 \$/MWh, 66 \$/MWh for a wind, solar, nuclear and coal power plant, respectively. For a more comprehensive overview from a hydropower perspective, see ¹¹⁷
<i>Cost of energy</i>	See section 2.
<i>Critical raw materials</i>	No, see section 4.
<i>Resource efficiency and recycling</i>	The lifespan of civil structures can reach 80 years if periodic maintenance is carried out, after that a retrofitting activity is required. Dams should be designed to withstand 1000 years floods. Electro-mechanical equipment has a lifespan 20-30 years. The overall efficiency (from water withdrawn to turbine release) is generally above 80% for large plants and above 65% for mini plants, and can reach 90%. Heat losses of generators can be recovered and used for heating. Reservoirs can have multiple purposes.
<i>Industry viability and expansion potential</i>	See section 4.
<i>Trade impacts</i>	See section 4. Huge impact, as hydropower embraces several engineering sectors.
<i>Market demand</i>	See section 3.
<i>Technology lock-in/innovation lock-out</i>	Several R&D activities are ongoing, although the main technology is well established.
<i>Tech-specific permitting requirements</i>	Several permitting procedures for land and water use ¹¹⁸ .
<i>Sustainability certification schemes</i>	The Hydropower Sustainability Alliance (HSA) created the Hydropower Sustainability Standard.
Social	
<i>S-LCA standard or best practice</i>	See the text.
<i>Health</i>	No direct emissions.
<i>Public acceptance</i>	Big projects may require reallocation of people changes in water resources management in the exploited river basin. On the other hand, there is often scarce awareness on the multi-purpose benefits of water reservoirs and hydropower schemes. Several initiatives for youth and women are ongoing in the hydropower sector.
<i>Education opportunities and needs</i>	Especially when hydropower is linked to the industrial heritage and cultural heritage (e.g. water mills) ^{119, 120} .
<i>Employment and conditions</i>	For employment data see section 0.
<i>Contribution to GDP</i>	See section 3.
<i>Rural development impact</i>	Especially in case of rural areas that host water mills and hydraulic infrastructures. 80% of the EU's installed capacity is in rural areas ¹²¹ .
<i>Industrial transition impact</i>	Hydropower can provide flexibility and can be hybridized with other energy technologies (e.g. floating photovoltaics, heat generation, batteries).
<i>Affordable energy access (SDG7)</i>	If well operated, hydropower can provide a better management of water resources and micro-hydropower can be installed in remote localities.
<i>Safety and (cyber)security</i>	Digitalisation is an emerging strategy to improve generation, extend lifespan and mitigate impacts, and thus cybersecurity is of high relevance. The EU is a lead exporter, thus the hydropower's manufacturing activity is a secure market for the EU, with no dependency from foreign countries.
<i>Energy security</i>	Water and energy storage. Energy can be stored in large-scale reservoirs and in PSH in larger quantities with respect to batteries. For example, the stored energy in the Blåsjø PSH reservoir, the Norway's largest reservoir (8 TWh), is equivalent to more than 40,000 times the Hornsdale battery park in Australia. In the EU, the largest reservoir for hydropower purpose hosts 3.12 TWh; it is in Spain and is PSH with turbine and pump installed capacity of 851 and 1184 MW, respectively (JRC hydropower database).

<i>Food security</i>	In the EU, 120 large reservoirs host hydropower and irrigation as first and second use, and 30 large reservoirs host irrigation and hydropower as first and second use, respectively, from ICOLD2023 database.
<i>Responsible material sourcing</i>	No.

Annex 2 Summary Table of Data Sources for the CETO Indicators

Theme	Indicator	Main data source
Technology maturity status, development and trends	Technology readiness level	Scientific literature
	Installed capacity & energy production	Eurostat, IHA, IRENA, IEA
	Technology costs	IEA, IRENA, ETIP Hydropower, IHA, Scientific literature
	Public and private RD&I funding	JRC
	Patenting trends	JRC, scientific papers
	Scientific publication trends	JRC, Scopus
	Assessment of R&I project developments	JRC, CORDIS
Value chain analysis	Turnover	JRC
	Gross Value Added	JRC
	Environmental and socio-economic sustainability	Scientific literature
	EU companies and roles	JRC, Hydropower and Dam website
	Employment	JRC, IRENA, EUROBSERVER
	Energy intensity and labour productivity	JRC
	EU industrial production	JRC
Global markets and EU positioning	Global market growth and relevant short-to-medium term projections	JRC, IEA
	EU market share vs third countries share, including EU market leaders and global market leaders	JRC
	EU trade (imports, exports) and trade balance	JRC
	Resource efficiency and dependencies (in relation EU competitiveness)	Scientific literature

Annex 3: details on investments

Private Equity refers to capital investments (ownership or interest) made into companies that are not publicly traded. Venture capital (VC) is a form of private equity and a type of financing that investors provide to start-up companies and small businesses that have long-term growth potential.

The early and later stages indicators that aggregate different types of equity investments in a selection of companies and along the different stages of their growth path. We only include pre-venture companies (that have received Angel or Seed funding, or are less than 2 years old and have not received funding) and venture capital companies (companies that have been part of the portfolio of a venture capital investment firm at some point).

The early stages indicator include Pre-Seed, Accelerator/Incubator, Angel, Seed and Early stage VCee investments; it also include public grants. At the time they raise such investments, those companies can usually be considered as start-ups. But while those companies often rely on innovative solutions and business models, such investments aim at financing the companies' operational expenditures and investment needs until they can scale their revenues and cannot be assimilated to R&I funding.

The later stages indicator reflect growth investments for the scale-up of start-ups or larger SMEs. It include Late Stage VCff, Small M&Agg and Private Equity Growth/Expansion. Very large early stage deals (outliers) are also re-classified as later-stage deals. Small M&A refers to the acquisition by an operating company of a non-control stake in a pre-venture or VC company. Later stages investments do not include: Buyout Private Equity and Public investments.

The lists of companies include two distinct populations: VC and corporate companies. Corporate companies is a selection of companies with a relevant patenting activity among the subsidiaries of top R&D investors from the EU Industrial R&D investment Scoreboard. VC companies are selected based on their activity description (specific keyword selection for each technology and expert inputs) and this selection does not rely on patents. This selection tries to focus on companies that develop and manufacture technological solutions as much as possible. It does not e.g. include operators, project developers, specific applications.

Those selections include all identified companies for each technology, irrespectively of their current operational status or of the fact that they have relevant investments or patenting activities over the current period. VC companies may e.g. currently be start-ups, may have been start-ups or larger SMEs that grew into larger private companies, went public or were acquired by larger companies. They may also currently be out of business.

As they focus on two specific populations, those lists only represent subsets of all market players in each value chains. The aim is however to illustrate the dynamics of emerging innovators with growth potential and large corporate innovators (that are responsible for most private R&I investments).

To support that analysis, the count of companies corresponds to the number of active companies over the current period. Active corporate companies have High Value Patents over the current period. Active VC companies either have been founded (irrespectively of received investments) or have received investments (irrespectively of their founding year) over the current period.

^{ee} Usually series A and B occurring within 5 years of the company's founding date

^{ff} Usually Series B to Series Z+ rounds and/or occurring more than 5 years after the company's founding date, as well as undisclosed series

^{gg} Small M&A refers to the acquisition by an operating company of a non-control stake

Annex 4: details on patents

Patent families (inventions) include all documents relevant to a distinct invention (e.g. applications to multiple authorities).

Statistics are produced based on applicants, considering applications to all offices and routes.

When more than one applicant or technology code is associated with an application, fractional counting is used to proportion effort between applicants or technological areas, thus preventing multiple counting. An invention is considered of high-value when it contains patent applications to more than one office.

Patent applications protected in a country different to the residence of the applicant are considered as international.

High-value considers EU countries separately, while for international inventions European countries are viewed as one macro category.

The CPC classification is not in use with the same degree of consistency across IPOs in Asia. The figures for the total number of inventions for Asian countries should be used with caution. This does not affect statistics for high-value and international inventions.

- Patent families (or inventions) measure the inventive activity. Patent families include all documents relevant to a distinct invention (e.g. applications to multiple authorities), thus preventing multiple counting. A fraction of the family is allocated to each applicant and relevant technology.
- High-value inventions (or high-value patent families) refer to patent families that include patent applications filed in more than one patent office.
- Granted patent families represent the share of granted applications in one family. The share is then associated to the fractional counts in the family.
- Flow of inventions (or destination of patent families) indicates where (in which national patent office) inventions are filed. This can be used to analyse the international flow of inventions.

Annex 5 Energy System Models and Scenarios: POTEnCIA and POLES-JRC

Annex 5.1 The POTEnCIA CETO 2025 Scenario

The *POTEnCIA CETO 2025 Scenario* has been generated with the Policy Oriented Tool for Energy and Climate Change Impact Assessment (POTEnCIA). Developed in-house by the European Commission's Joint Research Centre (JRC) to support EU policy analysis, POTEnCIA is an energy system simulation model designed to compare alternative pathways for the EU energy system. The core modelling approach of POTEnCIA (detailed in Mantzos et al., 2019) focuses on the economically-driven operation of energy markets and corresponding supply-demand interactions, based on a recursive dynamic partial equilibrium method.

The technology projections provided in the *POTEnCIA CETO 2025 Scenario* are obtained under a climate neutrality scenario aligned with the broad GHG reduction objectives of the European Green Deal. As such, this scenario reduces net EU GHG emissions by 55% by 2030 and 90% by 2040, both compared to 1990, and reaches net zero EU emissions by 2050. To model the uptake of different technologies under this decarbonisation trajectory, the scenario includes a representation at EU level of general climate and energy policies such as emissions pricing under the Emissions Trading System, as well as key policy instruments that have a crucial impact on the uptake of specific technologies. For instance, the 2030 energy consumption and renewable energy shares reflect the targets of the EU's Renewable Energy Directive (RED) and of the Energy Efficiency Directive (EED). Similarly, the adoption of alternative powertrains and fuels in transport is consistent with the updated CO₂ emission standards in road transport and with the targets of the ReFuelEU Aviation and FuelEU Maritime regulations.

Compared to the *POTEnCIA CETO 2024 Scenario* (Neuwahl et al, 2024), the *POTEnCIA CETO 2025 Scenario* incorporates many model enhancements and scenario-specific updates, most notably:

- The usage of the more recent JRC-IDEES 2023 data (Rózsai et al, 2025).
- Closer alignment to the National Energy and Climate Plans (NECPs) of the individual MS, which have been published in recent months.

A more detailed description of the *POTEnCIA CETO 2025 Scenario* will be available in the forthcoming report (Neuwahl et al., 2025).

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Annex 5.2 Global CETO 2°C Scenario 2025

The *Global CETO 2°C Scenario 2025* has been generated by the global energy model *POLES-JRC* (Prospective Outlook for the Long-term Energy System).

POLES-JRC covers the entire global energy system with a high level of regional detail, encompassing 66 countries and regions, from primary supply (fossil fuels, renewables) to transformation (power, biofuels, hydrogen, and hydrogen-derived fuels, such as e-fuels and ammonia) and final consumption sectoral demand (industry, buildings, transport). International markets and energy fuel prices are calculated endogenously. The model comprises a comprehensive portfolio of technologies, with technology dynamics and interactions across sectors modelled using endogenous technology learning.

Detailed documentation of the *POLES-JRC* model is provided in (Després et al., 2018). Techno-economic assumptions used in the current version of the model are provided in (Schmitz et al., 2025). The latter report provides also a comprehensive overview of the dynamics and interaction of various clean energy technologies until the end of the century. *POLES-JRC* results are published in the annual "Global Climate and Energy Outlooks" (GECO) report, which provides detailed country energy and GHG balances, and an online visualisation interface.

The *Global CETO 2°C scenario 2025* is designed to limit global temperature increase to 2°C at the end of the century. It is driven by a single global carbon price. The *Global CETO 2°C scenario 2025* builds on the *POLES-JRC* model version used for GECO 2024 (European Commission, 2025) and has been enhanced as following:

- Cost optimisation for electrolyzers powered by renewables (PV, wind) has been implemented considering an over-sizing of renewable capacities and battery storage.
- Electrolyser investment costs have been increased reflecting recent literature revisions.
- Updated investment costs for renewable technologies and utility battery according to (IRENA, 2025) as well as updated installed capacities for power generating technologies.
- Revised global wind profiles (off-shore and on-shore).
- Updated data on new vehicles and vehicle stock by transport mode (battery and fuel cell vehicles, ICE, hybrid) and vehicle type (passenger cars, light and heavy trucks, buses).
- Updated hydrogen infrastructure cost related to road transport.

POLES-JRC References

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List of abbreviations and definitions

AU = Australia

BR = Brasil

CA = Canada

CAPEX = capital expenditure

CF = capacity factor (-)

CH = Switzerland

EC = European Commission

EROI = Energy Return on Investment

EU = European Union EU27

GDP = Gross Domestic Production

GW = GigaWatt

ICOLD = International Commission on Large Dams

IEA = International Energy Agency

IHA = International Hydropower Association

IRENA = International Renewable Energy Agency

JP = Japan

JRC = Joint Research Centre

KR = Korea

LCOE= Levelized Cost of Electricity

MS = member state

MX = Mexico

NO = Norway

NZ = New Zealand

O&M = Operation and Maintenance

P = installed power (kW, GW)

PSH = pumped-storage hydropower

PV = photovoltaics

R&D = research and development

ROR = run of the river

RoW = Rest of the World

RSHP = reservoir storage hydropower plant

TR = Türkiye

TRL = Technology Readiness Level

TWh = TeraWatt per hour

U.K. = United Kingdom

U.S. = United States of America

WEFE = Water-Energy-Food-Ecosystem

y = year

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